

Pre-Grant Patents and Innovation Diffusion *

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Abstract

Are pre-grant patents effective forward-looking signals of global innovation diffusion? Patent grants are important legal milestones, but information diffusion about the underlying technologies occurs at the time patents are published. As the innovation literature rarely studies this distinction, I address this question using pre-grant patent flows across countries, sectors, and industries over time. This allows me to separately identify timing effects on diffusion and those that are running through either innovation or trade channels. At country level, results show that the innovation channel creates larger and more persistent total factor productivity (TFP) gains and stock price responses. This confirms that international technology spillovers originate from the expansion of the global stock of knowledge rather than strategic changes in trade intensity between home countries and their partners. A one-standard-deviation foreign pre-grant patent shock raises manufacturing-sector TFP by about 0.28%, and the R&D capital stock rises by roughly 0.18% with a one-year delay. This reflects their forward-looking nature, which prompts resource reallocation in anticipation of future productivity gains. These gains are especially pronounced in countries which manufacturing sectors that are more R&D-intensive. At the industry level, countries with more value-added-intensive industries are more capable of generating internationally influential frontier innovations and exhibit larger labour productivity gains.

Keywords: Pre-Grant patents; Innovation diffusion; Trade Exposure

JEL Codes: O33, O47, F14, F43, O31

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1 Introduction

When measuring the innovation and investigating its diffusion, the empirical literature has relied on post-grant patents to trace the global spread of technology (Jaffe et al., 1993; Bloom, Schankerman and Van Reenen, 2013; Keller, 2004; Peri, 2005). Yet this adoption overlooks a fundamental feature of the patents: the economic and informational content of an invention becomes publicly available when a patent is published. At that moment, citations begin, expectations adjust, and firms can already access the underlying knowledge (LaBelle et al., 2023). By the time a patent is granted which happens around two to five years later, most of the informational value has already diffused; granting serves a legal protection but adds little new scientific content and forward-looking signals.

At the same time, technology news literature (Beaudry and Portier, 2006; Barsky and Sims, 2011) treats anticipated technology signals country-specific and domestic. However, it fails to acknowledge that the expectations about future productivity not only depend on domestic innovation, but also on information about foreign technological advancement and their potential transmission across borders. Meanwhile, international R&D spillover literature emphasize that importing from technologically advanced economies facilitates learning-by-importing and embodied technology transfer (Coe and Helpman, 1995; Eaton and Kortum, 1999, 2002; Amiti and Konings, 2007). However, it does not look into how forward-looking it fails to identify whether international innovation diffusion is driven by foreign innovation itself or by changing trade patterns that shapes the exposure to foreign ideas. In addition, although absorptive capacity is known to determine how effectively economies internalise foreign knowledge (Cohen and Levinthal, 1989; Griffith, Redding and Van Reenen, 2004), we know little about whether early-stage patent signal such as pre-grant publications matter more for economies with stronger capital stock of R&D or more sophisticated industrial structures. Industry-level diffusion remains particularly underexplored in cross-country settings.

Against this background, this paper asks whether pre-grant patents serve as better forward-looking indicators of global innovation diffusion than granted patents and how do pre-grant patents facilitate the international diffusion across countries, sectors and industries.

At the country and sector level, the question is: How do countries respond when they are exposed to foreign pre-grant technological information, and which channel is the main driver for the propagation of

these early signals. To answer this question, I construct a pre-grant patent index that incorporate both foreign patent publication activities and trade to represent two channels - global innovation channel and trade channel. To reveal which channel is the main driver, I build up the index as the import-weighted sum of foreign patent publication activities while the global innovation channel fix the import weight while the trade channel fix the foreign innovation activities. I then use panel local projection to capture the effects of such forward-looking pre-grant patent shocks to domestic TFP and stock prices. I also compare the impulse responses functions of these key variables to those to granted patent shocks which built up in the same way. I found that firstly the global innovation channel is the main driver of pre-grant patent shocks diffusion across countries based on the fact that the countries have constant trade relations with its trade partners. Then the pre-grant patents shocks have caused larger responses in both domestic TFP and stock price. This result is not denying the importance of trade channel as the diffusion was mainly driven by global innovation channel but it is based on the constant and stable trade relations with trade partners. And we can also tell that we do not have to wait until the granted stage for patents to get notified and prepared in advance, as the patent publications represents the efforts of a country in new technology through research and development spending that represent the growing trend of technology for future production and productivity changes. Plus, patent publications represent the disclosure of information, which has initiated the diffusion earlier than granted patents.

I further extend this chapter to sectoral level analysis where I focus on traded sector since pre-grant patent index incorporated trade in its construction. I first construct a Bartik Shift Share instrument that fix the trade share and combine it with the shift of patent publication activities to guarantee the cleaner foreign technological influence at sectoral level, then I apply panel local projection to see the effects of sectoral pre-grant patent shocks over TFP and capital stock of R&D. This set up is in line with the existing literature of identifying external shifts (Bartik, 1991; Autor, Dorn & Hanson, 2013; Goldsmith-Pinkham, Sorkin & Swift, 2020). I found that both TFP and capital stock of R&D has shown such 1 year period of preparation following the sectoral pre-grant patent shocks, which confirms the fact that sectoral patent publications have given forward-looking signals for manufacturing sectors to prepare in advance so that they can absorb and gain from this diffusion better and earlier. These patterns suggest that pre-grant information enables manufacturing sectors to anticipate technological change and adjust research activities in advance. The results contribute to

the international R&D spillover literature by highlighting this preparation window and emphasizing the role of foreign-innovation shocks in driving cross-border productivity transmission (Keller, 2004; Peri, 2005). I also include sectoral absorptive capacity that indexed by R&D intensity to see if manufacturing sector across countries will exhibit different magnitude of productivity gains following this pre-grant patent shocks. The result shows that countries with more R&D intensive manufacturing sectors will be able to absorb more from this sectoral pre-grant patent shocks diffusion and have higher gains of TFP. This result highlights the importance of building up absorptive capacity through R&D in manufacturing sectors. Through targeted and higher volume of R&D, manufacturing sectors will be able to internalise and prepare through the patent publications activities from the manufacturing sectors of its trade partners to make the most of this sectoral pre-grant patent shocks. The idea is to get notified and put efforts to absorb through targeted R&D ahead of the actual introduction of new technology through patent publications rather than catching up after the arrival of new technology.

At the industry level, the question shifts from the receiving side of international technology diffusion toward the origin side of technological influence: after countries are exposed to and internalise frontier technological information, do they subsequently generate innovations that become internationally influential themselves? The motivation for the industry-level analysis comes directly from the behaviour of chemical and pharmaceutical industries within manufacturing sectors, where innovation diffusion relies heavily on early patent publications and citation networks rather than formal patent grants. In these industries, firms frequently build upon and cite previous patent publications long before technologies reach the granted stage, implying that technological diffusion and follow-on innovation begin during the pre-grant period itself.

To investigate this mechanism, I examine whether pre-grant technologies originating from specific countries and industries generate stronger downstream foreign citation responses and whether internationally influential innovations are associated with higher subsequent productivity gains. I employ LaBelle et al.'s (2023) quality-adjusted citation index to measure the downstream international influence of pre-grant patents. I then interact this citation-based measure with industry-level value-added intensity, which proxies industrial sophistication and production capability in line with the absorptive-capacity literature (Cohen & Levinthal, 1990; Griffith, Redding & Van Reenen, 2004). The results show that countries with more value-added intensive chemical and pharmaceutical industries generate

pre-grant technologies that receive stronger downstream foreign citation responses and are associated with larger labour productivity gains. These findings suggest that industrial sophistication strengthens the international influence and economic relevance of early-stage technological innovations.

Related Literature. This paper bridges and contribute to several strands of research—patent-based measures of technology shocks, international knowledge diffusion through trade, and absorptive capacity in shaping heterogeneous innovation uptake. Across all three, the contribution lies in showing that pre-grant patents provide earlier and more informative signals of global innovation diffusion than granted patents, which was a dimension that existing studies have largely overlooked.

Firstly, this paper connects and bridges two strands of the literature that have evolved in parallel over time. The first is the literature on news technology shocks (e.g. Beaudry and Portier, 2006; Barsky and Sims, 2011), which mainly studies how anticipation over future technological advancement would affect main macroeconomic variables and broader picture. While this literature emphasises how important the anticipation could be in the technology diffusion, it often observe the diffusion from more of the domestic perspective rather than the cross-border perspective of technology advancement. The second strand is the international R&D spillover literature (e.g. Coe and Helpman, 1995; Keller, 2004), which investigates how foreign innovation affects domestic productivity through trade linkages, but treats diffusion as a largely backward-looking and contemporaneous process. This paper integrates these two perspectives by showing that international R&D spillovers can operate as a source of global technology news.

Then, this paper re-evaluates and find the gap the related work that uses patents to measure technological progress, where they typically rely on post-grant measures such as granted patent counts, forward citations, and R&D expenditures to capture realised innovation (Jaffe et al., 1993; Hall, Jaffe and Trajtenberg, 2005; Bloom et al., 2013). More recent contributions introduce patent-based news shocks as forward-looking indicators of innovation, though these have been studied almost exclusively in single-country contexts, especially for the United States (Beaudry and Portier, 2006; Kogan et al., 2023; Cascaldi-Garcia and Vukotić, 2023). These studies implicitly assume that economically relevant knowledge diffuses after the grant stage. In contrast, recent evidence in LaBelle et al. (2023) shows that the key informational content of a patent is disclosed at publication which takes place roughly eighteen months after filing when technical specifications become publicly available and citations first

begin. Firms effectively access and respond to new knowledge at this publication stage, rather than waiting for legal certification.

Building on this insight, the present paper systematically compares pre-grant and granted patent measures across multiple countries, sectors, and industries. The results show that pre-grant patents generate larger, earlier, and more persistent responses in TFP and higher impact responses of stock prices. This timing of positive responses from patent publication contrasts those of granted patents reveals the efficient innovation diffusion by pre-grant patents: by the time patents are granted, much of the relevant knowledge has already diffused.

Another strand of literature examines how international productivity spillovers happen through trade and global openness. Literature shows that importing from technologically advanced partners facilitates learning-by-importing and embodied technology transfer (Coe & Helpman, 1995; Eaton & Kortum, 1999, 2002; Amiti & Konings, 2007; Bernard et al., 2007), and recent models emphasise how supply-chain linkages transmit productivity shocks across borders (Buera & Oberfield, 2020). Yet almost all of this literature uses realised innovation—granted patents, R&D stocks, or aggregate productivity in trade partners to proxy for foreign knowledge. These measures capture ex-post technological achievement rather than early-stage information.

By contrast, this paper isolates two structural diffusion channels: an innovation-driven channel, where foreign knowledge evolves while trade exposure is held fixed; and a trade-driven channel, where foreign innovation is fixed and only import intensity varies. In this framework, trade exposure determines which foreign ideas a domestic economy has access to, while pre-grant publications determine when those ideas become visible. Using pre-grant patents in this context therefore refines the mechanism by capturing early signals that align naturally with how diffusion operates through global production networks. The empirical findings clearly show that the innovation-driven channel dominates. Changes in global knowledge—measured using pre-grant patents—produce strong and persistent spillovers, whereas shifts in import patterns alone produce weak or statistically insignificant effects. This contrast suggests that the primary driver of international diffusion is movement at the knowledge frontier, rather than contemporaneous changes in trade composition. In this paper, trade does not “cause” diffusion by itself; rather, import patterns capture the intensity of exposure to foreign knowledge embedded in traded goods, suppliers, and global value chains. The paper also extends

the international R&D spillover literature by incorporating publication-stage patent information and showing that diffusion begins earlier than standard trade-based models imply.

Lastly, studies on how domestic absorptive capacity shapes economies' ability to internalise new technologies emphasises the importance of R&D intensity, human capital, and organisational capability (Cohen and Levinthal, 1990; Griffith, Redding and Van Reenen, 2004). Complementary work documents persistent heterogeneity in technology adoption across countries and industries (Comin, Hobijn and Rovito, 2006; Comin and Mestieri, 2018). This paper contributes in two ways. At the sector level, it shows that countries with more R&D-intensive manufacturing sectors exhibit substantially larger productivity responses to pre-grant patent shocks. This provides new evidence that absorptive capacity matters already at the publication stage: early stage knowledge is valuable only when domestic sectors have the capability to process, translate and integrate it.

Finally, forward citations to pre-grant patents provide a direct measure of how quickly and intensively new ideas are taken up by subsequent innovators. The analogy to academic research is exact: scholars do not wait for journal publication to cite and build upon a working paper. Some working papers might never get to the publication stage or take such a long time (could be even after 6 years) yet still at the stage of working paper, yet it has been cited in new papers which symbolises the early stage of knowledge diffusion. Similarly, firms do not wait for legal grant to build upon a disclosed patent. The citation literature relies on this mechanism to trace knowledge spillovers (Jaffe et al., 1993; Hall, Jaffe & Trajtenberg, 2005; Bloom, Schankerman & Van Reenen, 2013), and LaBelle et al. (2023) show that citations received shortly after publication capture the first wave of technological uptake. In this sense, pre-grant publications contain the signal of the initial codified disclosure, whereas early citations reveal the early internalisation of that signal. At the industry level, the paper uses quality-adjusted citation index by LaBelle et al (2023) to demonstrate that countries with higher value-added intensive chemical industries can better internalise the citation of patents into actual labour productivity gains. This results have reflected the importance of industrial sophistication that enables industries in uptaking the early stage patents through citation and transfer them into the gains.

There is a practical policy motivation as well. Multiple reports highlight the widening “diffusion gap” between frontier firms and laggard sectors, emphasising that technology shocks often reach domestic industries suddenly and unevenly through global supply chains (OECD, 2015; OECD, 2019;

IMF, 2021). Without forward-looking indicators, domestic sectors adjust only after technologies are embodied in imported goods, by which time adaptation is costlier and slower. Pre-grant publications provide precisely the kind of anticipatory information that industrial-policy frameworks seek: early signals that allow firms and governments to prepare workforce skills, research programmes, and industrial upgrading before new technologies fully materialise in the domestic production networks.

2 Data

2.1 Country level data and classification

At the country level, I employ pre-grant patent data from the INPACT-S database constructed by LaBelle et al. (2023). This dataset aggregates patent applications by private firms across a country’s trade partners across three major international jurisdictions: the United States Patent and Trademark Office (USPTO), the European Patent Office (EPO), and the Patent Cooperation Treaty (PCT). Because direct and consistent data on raw filing dates are often fragmented or incomplete across global jurisdictions, LaBelle et al. (2023) reconstruct the application series retrospectively by tracing realized public patent publications back to their underlying filing origins to recover fractional country ownership. While the original data creators frame this series around initial filings, the retrospective construction means the variable practically isolates the specific subset of patent applications that successfully transitioned to the formal publication stage. Because my core empirical focus is on open-economy innovation diffusion—a mechanism that structurally requires public information disclosure—I explicitly refer to this fractionally assigned series as patent publications throughout this paper to align the data definition with the underlying economic transmission channel

For the purposes of this thesis, this data construction strategy offers a distinct empirical and conceptual advantage for studying open-economy innovation diffusion. By identifying applications exclusively through subsequent publication records, the sample systematically filters out early rejections, administrative withdrawals, and abandoned ‘ghost’ filings that are legally sealed and never enter the public domain. Consequently, this variable captures the exact subset of patent applications that successfully reached the formal publication stage. While these observations are indexed by their his-

torical application footprints to isolate the precise timing of knowledge production, they represent a high-quality, verified baseline of viable technological information that is guaranteed to cross the 18-month statutory confidentiality window and achieve public visibility—making them the ideal proxy for tracking the origins of diffusable global knowledge.

I selected 21 major advanced and emerging economies that have been outstanding in either innovation or trade in the past several decades. Firstly it is the 17 OECD countries that reflects the core of global innovation and high absorptive capacity with their continuous input in R&D that consist of the United States, Canada, Japan, South Korea, Australia, Singapore, Germany, France, the United Kingdom, Italy, the Netherlands, Sweden, Denmark, Finland, and Switzerland. Switzerland is selected due to its high level of patent intensity, R&D spending and cross-border collaboration that makes it stand out in innovation index such as Global Innovation Index (GII).

Meanwhile, several emerging economies have been playing an important role in international trade and innovation to grow rapidly over time. I selected four BRICS economies: China, India, Brazil and South Africa. The inclusion of these emerging economies has made the contribution novel as not many analysis has been discussing the role of emerging economies in the global innovation diffusion. Meanwhile, these countries have increased their integration into global supply chains through both increasing participation of international trade and patenting activities. Their growing importance in global patenting and trade activities have also changed the geography of innovation and trade.

Therefore, the combination of advanced and emerging economies in the analysis will provide a more realistic and comprehensive view of how pre-grant patents could give forward-looking signals on domestic economies through international trade.

The inclusion of emerging economies and high-performing smaller developed economies allows for a more realistic and comprehensive view of how pre-grant innovation signals propagate through international trade. This extended coverage reflects the increasingly multi-polar nature of global innovation and ensures that the empirical strategy accounts for the complexity of modern technology diffusion.

In this chapter, the pre-grant patents are defined as the patent publication that typically occurs around 18 months after filing, although the exact timing may vary across countries and patent characteristics. In particular, some might question that the institutional changes in some jurisdictions such as the United States around 2000 affected publication practices. To answer this question, it is crucial

to emphasize that the empirical strategy in this paper does not rely on an exact publication lag, but rather on the fact that patent applications become publicly observable well before grant decisions and before the associated productivity gains materialise. Therefore, the 18 months is an average time gap that symbolise the line for patents to disclose the scientific information that shapes the anticipation of the countries and sectors.

2.2 Sector level data and classification

At the sector level, pre-grant patent data spans from 2000 to 2014 (patent publications) are sourced from the Final Report on the Collection of Patents and Business Indicators by Economic Sector: Societal Grand Challenges and Key Enabling Technologies (European Commission, 2021), where filings are categorized by NACE Rev.2 sector codes. The sectoral pre-grant patent index is constructed from patent publications by private firms only. This choice emphasizes commercially-oriented innovation that is more likely to influence firms’ production decisions and to be transmitted internationally via trade. Public-sector or university publications are excluded to sharpen the focus on market-driven technological activity. Due to data availability constraints, the sectoral analysis is restricted to OECD economies, as comparable industry-level measures are not available for BRICS countries over the sample period.

In this chapter, I follow the classification practice of Cai, Li, and Santacreu (2022) to take manufacturing sectors (NACE 10-32) as traded sectors. Manufacturing sectors include fabricated metao products, chemicals, machinery and textiles that usually exhibit high level and volume of innovation through cross-country trading activities. Meanwhile, this classification reflects the exposure of manufacturing sectors to foreign innovation shocks through imported inpus, capital equipments and intermediate goods. NACE manufacturing sector is mapped with with the sectorl TFP data from EU KLEMS for the followig analysis and pre-grant patent index constructions. The classification adopted in this paper is summarized in Table 1.

Table 1: Sector Classification by Patent and TFP Codes

Sector Patent Code	Sector TFP Code	Sector Description	Classification
NACES 10–32	ISIC D	Manufacturing Sectors	Traded

Why Focus on Manufacturing Sectors

The analysis focuses exclusively on manufacturing sectors because the country level empirical strategy identifies imports as the channel for foreign innovation shocks, therefore it is essential to concentrate on sectors that are highly integrated into international trade. Manufacturing sector accounts for the vast majority of cross-border flows in technology that are embodied goods, which makes them the most relevant domain for studying trade-mediated diffusion of pre-grant patent signals. Some industries such as machinery, electronics and chemicals are active in patenting activities due to its high R&D intensity and its capability of transferring R&D output into actual production, which provides them a good position in global value and supply chain. Meanwhile, existing empirical evidence (Keller, 2004) show that around 90% of the international R&D spillover are transmitted through the trade in manufacturing sector. Coe, Helpman, and Hoffmaister (2009), along with Papageorgiou and Spatafora (2012), show that productivity gains from foreign innovation are concentrated in tradable manufacturing rather than in services.

Limiting the analysis to manufacturing therefore ensures theoretical consistency with the diffusion mechanism under study and empirical alignment with where spillovers are most observable. Manufacturing represents the most traded, most R&D-intensive, and most innovation-responsive part of the economy—making it the ideal setting for analysing the transmission of pre-grant patent shocks through international trade.

2.3 Industry level data and classification

I use the quality-adjusted citation index from LaBelle et al. (2023) at industry level as they weight patent families by their early citation to better reflect the quality of patent publications. This index captures both volume and the relevance of pre-grant patent innovations that provide a more informative forward-looking signal of cross-industry knowledge spillovers.

To classify industries within the manufacturing sector, I rely on the International Standard Industrial Classification (ISIC Rev.3) at the 2-digit level. This framework provides a consistent mapping of industries across countries as shown in table 3.

Table 2: Selected ISIC Rev.3 (2-digit) Manufacturing Industry Codes

Code	Industry Description
24	Chemicals and chemical products

3 Country Level Empirical Strategy & Results

Pre-grant patents offer a uniquely informative and economically meaningful measure of technological innovation because they capture the moment when new knowledge is first disclosed to the world. At publication—typically eighteen months after filing—the full technical content becomes public, citations begin, and firms adjust expectations, R&D portfolios, and sourcing strategies in response. Although only a subset of published applications is ultimately granted, many non-grants reflect prosecution costs, strategic withdrawals, or portfolio reorganisation rather than a lack of technical substance or quality. Importantly, the knowledge disclosed at publication is already citable which reflects the diffusion of new ideas and knowledge. In this sense, the publication stage rather than the grant stage, marks the actual beginning of knowledge diffusion. In policy practice, reports emphasise that global technologies often arrive suddenly through supply chains, which does not offer enough space for domestic firms to react and reallocate resources for catching up with the foreign productivity changes. This widens the productivity and innovation gaps between frontier and laggard sectors (OECD, 2015; IMF, 2021). This gap between frontier and lagged sectors have made forward-looking signals necessary for timely adjustment. Early forward citations to pre-grant publications further reveal how rapidly and intensively new ideas are internalised, mirroring the way academic working papers diffuse before journal publication (Jaffe et al., 1993; Bloom, Schankerman & Van Reenen, 2013). Therefore, combining pre-grant patent publications, trade exposure, and early patent citations provides a conceptually coherent and empirically powerful way to trace anticipation, transmission, and absorption in global technology diffusion which reveal the earlier timing of forward-looking dynamics that granted-patent fail to capture.

Apart from the objective of investigating whether pre-grant patents works better than granted patents in forward-looking signal for future innovation diffusion. It is also important to understand through which channel does innovation diffuse from foreign countries to domestic economies. To

address this, I construct a two-channel framework: (i) the foreign innovation-driven channel, where trade linkages are held fixed to identify the effects of evolving foreign innovation activity, and (ii) the trade-driven channel, where global innovation is held fixed to identify the impact of changing trade integration on knowledge transmission. This design offers a novel way to disentangle the main driver of international technological diffusion. By using pre-grant patents as forward-looking indicators that contains the information disclosure of foreign new knowledge, this approach provides an early and exogenous measure of global innovation dynamics that further contribute to the existing studies that rely on granted patents.

3.1 Aggregate TFP under Global Innovation Channel: Pre-Grant versus Granted Patent Shocks

$$Z_{i,t}^{\text{Innovation}} = \sum_{k \neq i} \omega_{ik}^{\text{Base}} \cdot \text{PreGrant}_{k,t}, \quad \omega_{ik}^{\text{Base}} = \frac{1}{5} \sum_{s=1981}^{1985} \frac{\text{Imports}_{i \leftarrow k,s}}{\sum_{k' \neq i} \text{Imports}_{i \leftarrow k',s}}. \quad (1)$$

$Z_{i,t}^{\text{Innovation}}$ measures each domestic country i 's exposure to exogenous foreign innovation shocks from trade partners k , with import shares $\omega_{ik}^{\text{Base}}$ fixed at their 1981–1985 averages. This ensures that the structure of trade linkages remains constant, so that all variation arises from changes in partner-country pre-grant patenting activity ($\text{PreGrant}_{k,t}$). These patent publications represent the earliest observable stage of information disclosure of the foreign patents.

The intuition of fixing the import weights for corresponding foreign patent publications is to investigate if countries will be able to benefit from its trade partners' increasing volume of patent activities based on their constant and stable trade relations. This captures the pure foreign innovation activities that are exogenous of a home country's own economic fluctuations

$$\Delta \log TFP_{i,t+h} = \alpha_i + \delta_t + \beta_h Z_{i,t}^{\text{Innovation}} + \Gamma_h' W_{i,t-1} + \varepsilon_{i,t+h}, \quad h = 0, \dots, H, \quad (2)$$

In equation 2, h denotes the projection horizon in the local projection framework, so that β_h captures the response of $\Delta \log TFP$ at horizon h following the innovation shock. The control vector $W_{i,t-1}$ includes lagged TFP dynamics and lagged macroeconomic controls (e.g. GDP growth), all dated

$t - 1$ to ensure they are predetermined. For each horizon $h = 0, \dots, H$, the regression is estimated separately.

To validate that pre-grant patents give more forward-looking signals than granted patents, I also construct the index in the same way but replace the foreign patent publications with the granted patents from trade partners. While the granted patents have been the centre of patent-based literature (Jaffe et al., 1993; Keller, 2004; Bloom, Schankerman & Van Reenen, 2013) due to its stability and full legal protection, yet they reflect ex-post technological outcomes rather than revealing the timing of when knowledge actually begins to diffuse. Constructing the similar index with granted patents provide a parallel set of benchmark results to confirm if patent publications have really sigified domestic economies earlier than granted patents. Indeed, since publication is when firms first observe codified knowledge and begin citing, experimenting, and adjusting expectations, stronger anticipatory responses to pre-grant shocks would confirm that early disclosure of patent information drives global technology spillovers and we do not need to wait until it is granted, which the process is usually time- and money consuming with a lot of uncertainty. This comparative exercise thus directly tests whether pre-grant patents truly contain unique forward-looking information that granted patents cannot provide.

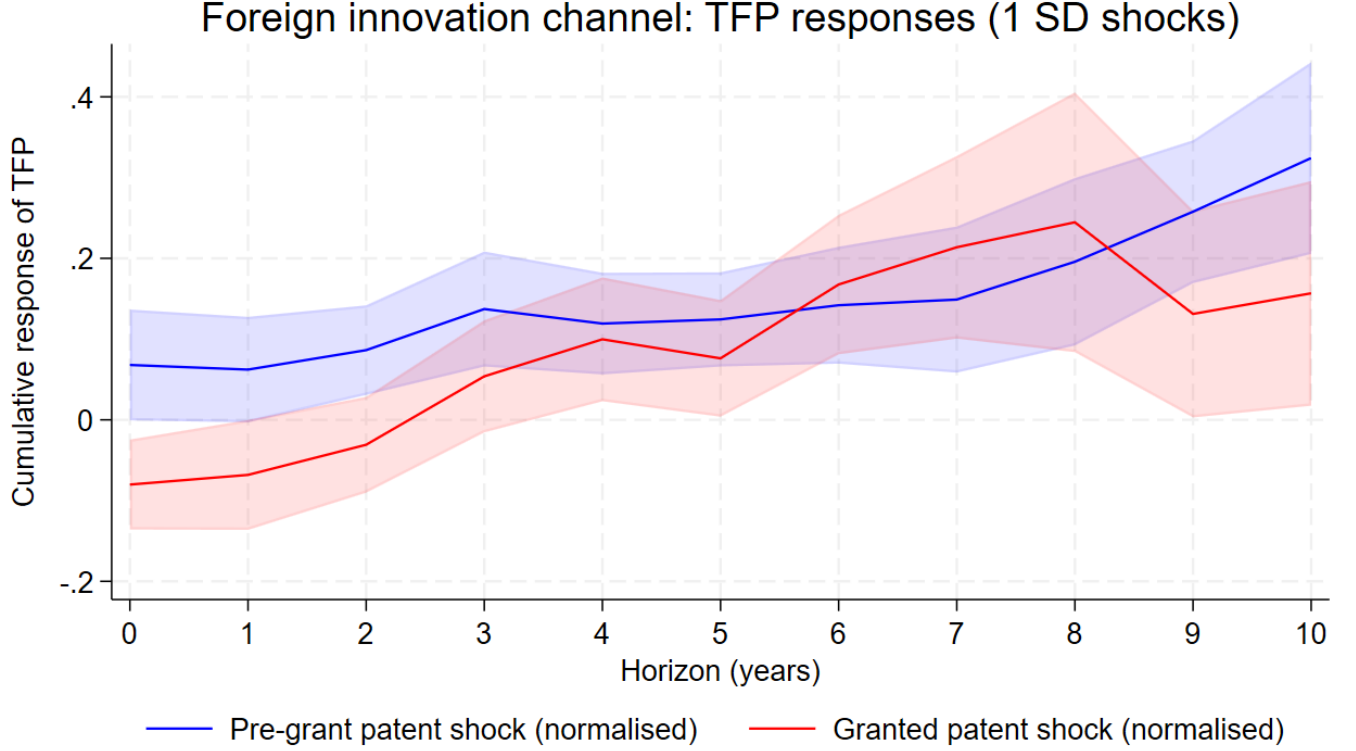


Figure 1: TFP Response to Pre-Grant Patent Shocks (Fixed Trade Share)

Notes: This figure plots the cumulative impulse response of domestic TFP (in percentage points) to a one standard deviation increase in the foreign pre-grant patent index. Shaded areas denote 68% confidence intervals constructed using standard errors clustered at the country level.

In figure 1, positive pre-grant patent shocks generate an larger increase in TFP both on impact and in the long run than that to granted patent shocks. The long-term build up in TFP fits with existing literature (Aghion & Howitt 1992; Aghion et al. 2019) where early knowledge spillovers expand the global technological frontier and triggers further innovation and productivity upgrades over time. This result confirms that early-stage foreign innovation measured by patent publications. This is in line with the existing literature of news shocks that technological news produces forward-looking immediate responses in key economic variables (Beaudry & Portier 2006; Barsky & Sims 2011; Atalay et al. 2020). Meanwhile, the result extend this claim further under the global innovation channel and validate the economic meaning of patent publications as the main driver of international technology diffusion. This set of benchmark result also provides the first cross-country evidence that early patent disclosures matter more than grants for productivity dynamics.

Thus, it is valid to conclude that the diffusion of the global innovation at granted patent stage has

died out, and the effect is much weaker because firms have already reacted to the relevant information at the earlier patent publication stage. This is consistent with the idea that granted-patent shocks are closer to realised technology shocks, not news shocks, echoing the distinction made in Azzimonti et al (2018). and in Aghion (1992)’s work on innovation cycles, where timing of disclosure determines the strength of responses. Plus, this muted result mirrors existing international R&D-spillover findings (Keller, 2004; Bloom et al., 2013).

In conclusion, the pre-grant patents works better than granted patents under the global innovation channel, meaning that we do not have to wait until the granted patent stage to adjust and prepare for the arrival of future technologies, and we get to gain more by monitoring the changes of foreign patent activities. Since publication means the moment an invention is published, its full technical content becomes publicly observable. Firms abroad can update expectations, initiate reorganisation of production plans, and adjust R&D direction. This mechanism aligns with the role of news shocks in macroeconomics, where forward-looking agents respond to information before the technology itself diffuses, which is consistent with Azzimonti, Francis & Rousakis (2018) and related news-shock literature.

3.2 Aggregate TFP under the Trade Channel: Pre-Grant versus Granted Patents

$$Z_{i,t}^{\text{Trade}} = \sum_{k \neq i} \omega_{ik,t} \cdot \text{Pre}\bar{\text{Grant}}_k, \quad \text{Pre}\bar{\text{Grant}}_k = \frac{1}{5} \sum_{s=1981}^{1985} \text{PreGrant}_{k,s}. \quad (3)$$

In the trade-driven channel, I hold the foreign patent publication constant by fixing each partner’s patent publication at its 1981-1985 average ($\text{Pre}\bar{\text{Grant}}_k$), while allowing import shares $\omega_{ik,t}$ to vary over time. This identifies the trade-driven channel to see if the changing trade integration could increase a country’s productivity gains from a constant international knowledge pool presented by pre-grant patents. The variation in $Z_{i,t}^{\text{Trade}}$ therefore captures how traded participation in global value chains influences the extent of technology diffusion.

This specification tests whether countries that strengthen trade ties with innovation-intensive partners gain more from this international technology diffusion. In the trade-driven channel, the idea is that trade patterns shape a country’s exposure to foreign technologies. This empirical setting is con-

sistent with the literature on technology transfer through trade (Coe & Helpman, 1995; Keller, 2004; Eaton & Kortum, 1999, 2002), where changes in import shares reflect shifts in domestic production and participation in global supply chain, which further influence the access to foreign knowledge. Therefore, this setting clarifies that trade-driven variation captures adjustments in exposure and provides a complementary benchmark against the global innovation channel in the level of diffusion.

I then employ panel local projection to capture dynamic responses of the TFP and stock price too (Jordà, 2005). The baseline equations are given by:

$$\Delta \log TFP_{i,t+h} = \alpha_i + \delta_t + \beta_h Z_{i,t}^{\text{Trade}} + \Gamma'_h W_{i,t-1} + \varepsilon_{i,t+h}, \quad h = 0, \dots, H. \quad (4)$$

Here, $x_{i,t+h}$ denotes either total factor productivity (TFP) or real stock prices. Country fixed effects (α_i) control for structural heterogeneity, year fixed effects (δ_t) absorb global shocks, and lagged controls ($W_{i,t-1}$) such GDP and past productivity. The sequence of coefficients β_h traces the dynamic propagation of innovation shocks across time horizons h under the trade-driven channel.

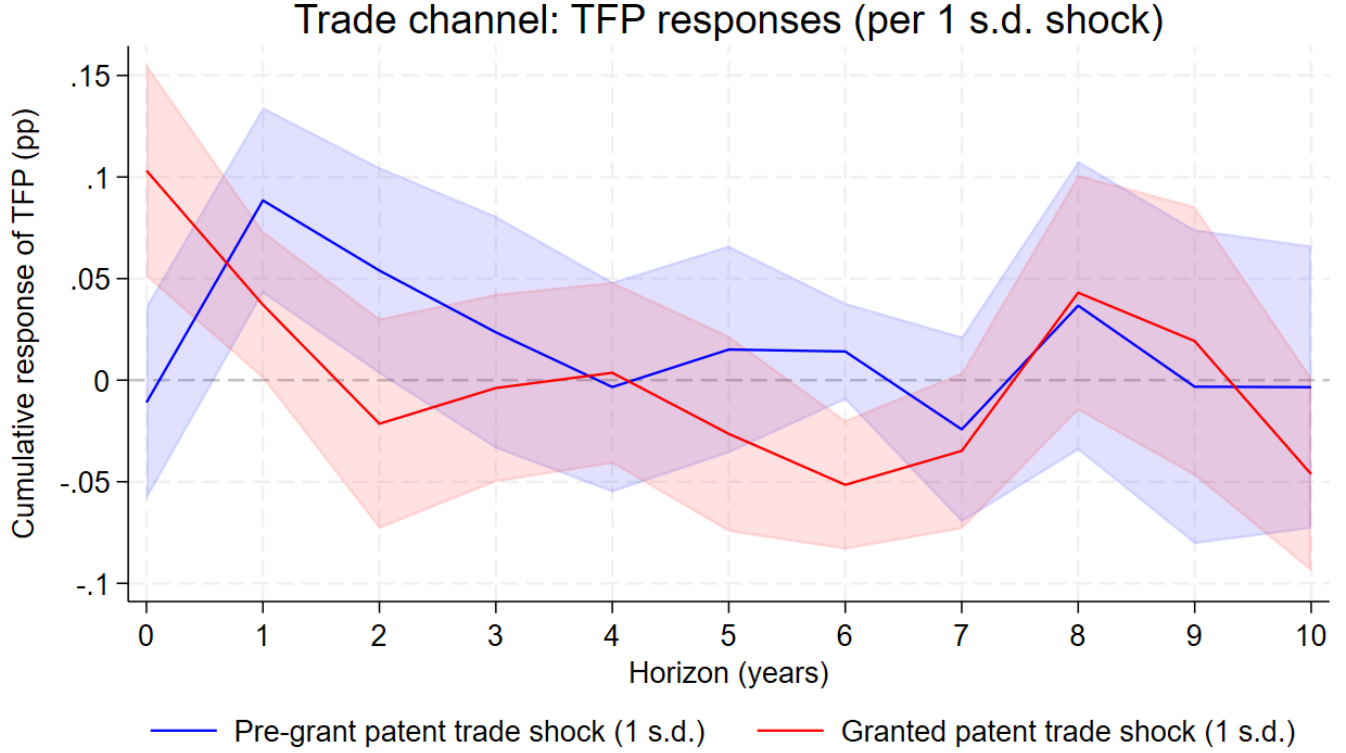


Figure 2: TFP Response to Pre-Grant Patent Shocks (Fixed Innovation Source)

Notes: This figure plots the cumulative impulse response of domestic TFP (in percentage points) to a one standard deviation increase in the foreign pre-grant patent index through the trade channel. Shaded areas denote 68% confidence intervals constructed using standard errors clustered at the country level.

The response of TFP to granted patent shocks in the trade-driven channel is small, negative, and statistically weak. The gradual decline of TFP to pre-grant patent shocks is consistent with trade-reallocation adjustment costs that is in line with Autor, Dorn & Hanson (2013) and Bloom et al. (2016), where shifts in import composition temporarily reduce productivity as sectors reorganize. Then the variation of trade relations itself is not a driver of diffusion, as it does not generate knowledge spillovers without the actual foreign patent activities. Compared to ther results under global innovation channel, we can tell that the changes in foreign patenting activities through patent publication is the key driver of cross country technology diffusion.

3.3 Stock Price

To further validate whether foreign innovation propagates from the anticipatory patent publications rather than changes in trade relations, I replace TFP with stock price in the panel local projections under the same two-channel framework. The objective is to see if the responses of stock price across countries show the similar trend to TFP and if there is the differentiation of responses through different channels. In line with Beaudry and Portier (2006), stock price is a good indicator for this analysis as it provides a direct evidence of whether financial markets react to the disclosure of information about future technological diffusion rather than legal protection of patents or changes in trade exposure.

Setting 1: Foreign Innovation Channel Diffusion (Fixed Trade Channel)

$$Z_{i,t}^{\text{Innovation}} = \sum_{k \neq i} \omega_{ik}^{\text{Base}} \cdot \text{PreGrant}_{k,t}, \quad \omega_{ik}^{\text{Base}} = \frac{1}{5} \sum_{s=1981}^{1985} \frac{\text{Imports}_{i \leftarrow k,s}}{\sum_{k' \neq i} \text{Imports}_{i \leftarrow k',s}}. \quad (5)$$

$$\Delta \log \text{StockPrice}_{i,t+h} = \alpha_i + \delta_t + \beta_h Z_{i,t}^{\text{Innovation}} + \Gamma'_h W_{i,t-1} + \varepsilon_{i,t+h}, \quad h = 0, \dots, H, \quad (6)$$

Results - Innovation Channel

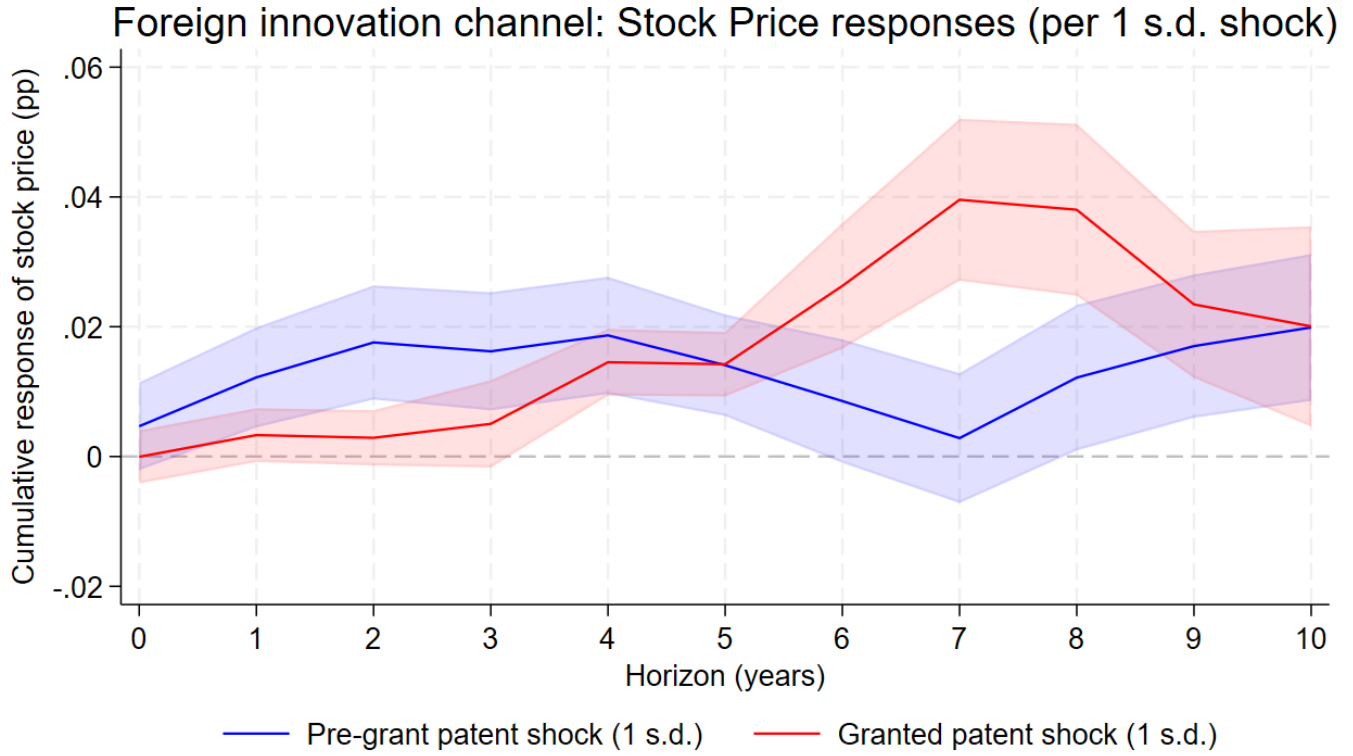


Figure 3: Stock Price Responses to Pre-Grant and Granted Patent Shocks (Foreign Innovation Channel)

Notes: This figure plots the cumulative impulse response of domestic stock prices (in percentage points) to a one standard deviation increase in the foreign pre-grant and granted patent index under the foreign innovation channel. Shaded areas denote 68% confidence intervals constructed using standard errors clustered at the country level.

In figure 4, stock price has higher responses to pre-grant patent shocks than that to granted patent shocks. Such result has confirmed the forward-looking nature of pre-grant patents as innovation signals better than granted patents. The responses of stock prices to granted shocks are so small that we can tell the diffusion has almost died out at granted stage. This is in line with the news shocks literature (Beaudry & Portier, 2006; Barsky & Sims, 2011; Angeletos et al., 2020) where financial market sensitively react to disclosure of information about future productivity changes before the real economic adjustments occur, and the pre-grant patents have extended it a bit further that the disclosure of information by pre-grant patents serves a better forward-looking signal of future technological changes than granted patents.

The stronger responses of stock price to pre-grant patent shocks reflect investors' interpretation of patent publication as credible news about future technological changes. And the results validates

the insights of Aghion et al. (2009) and Acemoglu et al. (2018) that financial markets will act at the early stage of innovation-driven growth. Since this chapter bring the analysis to the international context, pre-grant patents trigger cross-country expectations because the patent publications as pre-grant patents have created new and authentic access to all potential stakeholders for forward citations.

Setting 2: Trade Channel Diffusion (Fixed Global Innovation Channel)

$$Z_{i,t}^{\text{Trade}} = \sum_{k \neq i} \omega_{ik,t} \cdot \text{Pre}\bar{\text{Grant}}_k, \quad \text{Pre}\bar{\text{Grant}}_k = \frac{1}{5} \sum_{s=1981}^{1985} \text{PreGrant}_{k,s}. \quad (7)$$

$$\Delta \log \text{StockPrice}_{i,t+h} = \alpha_i + \delta_t + \beta_h Z_{i,t}^{\text{Trade}} + \Gamma'_h W_{i,t-1} + \varepsilon_{i,t+h}, \quad h = 0, \dots, H. \quad (8)$$

Trade Channel Results

Results - Trade Channel

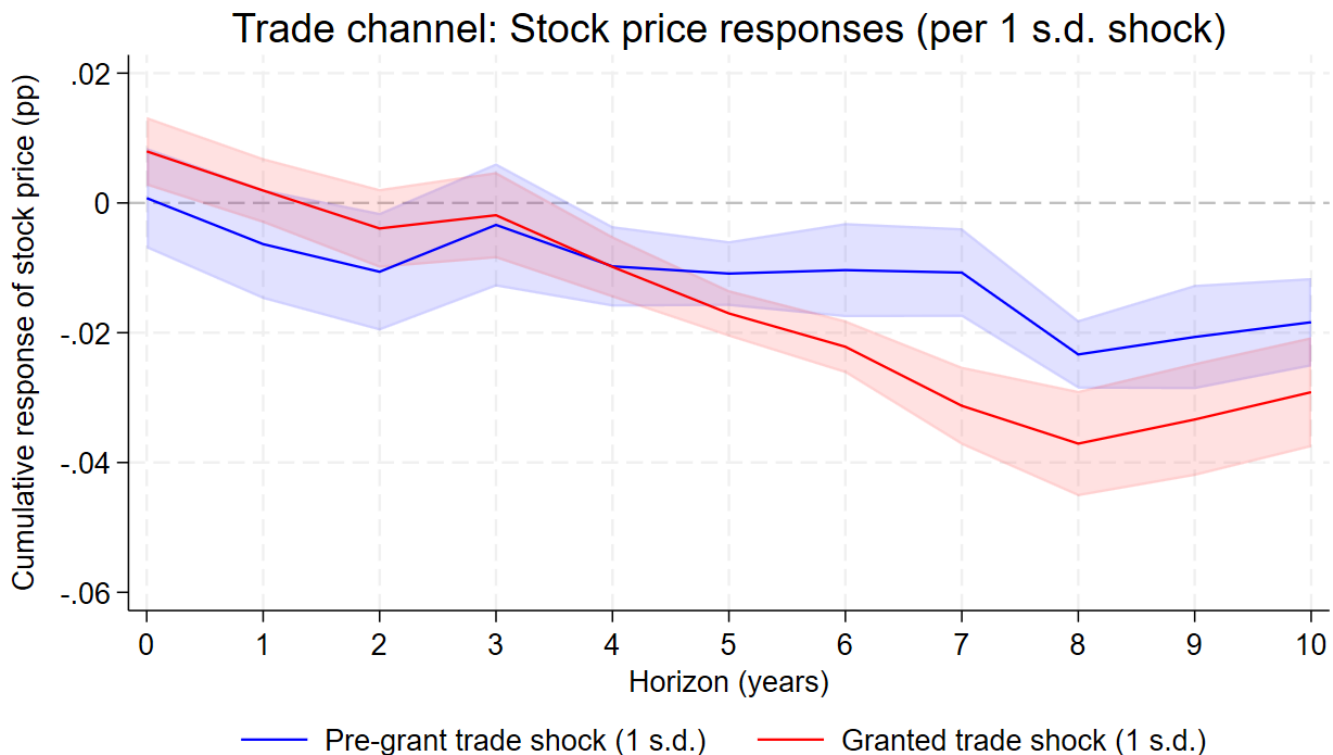


Figure 4: Stock Price Responses to Pre-Grant and Granted Patent Shocks (Trade Channel)

Notes: This figure plots the cumulative impulse response of domestic stock prices (in percentage points) to a one standard deviation increase in the foreign pre-grant and granted patent index under the trade channel. Shaded areas denote 68% confidence intervals constructed using standard errors clustered at the country level.

The responses of stock price to both pre-grant and granted patents under the trade channel are both weaker than those under the innovation channel. This is consistent with the existing literature that financial markets are sensitive to information shocks rather than shifts in trade relations (Forbes & Rigobon, 2002; Goldberg & Campa, 2010). The muted responses support the interpretation that trade serves as an exposure rather than a source of international technology diffusion.

This analysis documents cumulative impulse responses for stock price to capture the long-run valuation effects of pre-grant patent shocks. Cumulative responses can summarise the persistent revision in expectations triggered by disclosure of information by patent publication about future technological changes. Meanwhile, stock prices incorporate information quickly in the shorter horizons so the responses will be taken as the complementary to just test the driver of international technology diffusion.

4 Sectoral Level Empirical Strategy & Results

At the sector level, the goal is to construct a cleaner measure of foreign early-stage innovation exposure for each domestic manufacturing sector. The country-level results have shown that international diffusion is driven almost entirely by the innovation-driven channel—that is, by changes in foreign knowledge rather than by shifts in import composition. Therefore, this motivates the empirical strategy at sector level moving to a Bartik-style design, which isolates exogenous foreign-sector shocks while holding domestic trade exposure fixed, consistent with the shift–share logic used in Autor, Dorn and Hanson (2013) and formalised in Goldsmith-Pinkham, Sorkin and Swift (2020).

Before instrumenting, the baseline exposure of country i and sector s to foreign pre-grant innovation can be written as a weighted sum of partner-sector pre-grant patents:

$$\text{SectorPreGrant}_{i,s,t} = \sum_{k \in \mathcal{K}_i, k \neq i} w_{iks,t} \cdot \text{PreGrant}_{k,s,t},$$

where $w_{iks,t}$ denotes time-varying import shares capturing the contemporaneous trade exposure of sector s in country i to sector s in partner country k , and $\text{PreGrant}_{k,s,t}$ is the volume of pre-grant patent publications in sector s of trade partner k at time t .

Because $w_{iks,t}$ evolves with domestic trade choices, business cycles, or sectoral reallocation, $\text{SectorPreGrant}_{i,s,t}$ conflates foreign technological shocks with endogeneity.

To isolate the exogenous variation coming from foreign innovation, I construct a Bartik shift–share instrument starting from the sector-specific foreign shocks based on pre-grant patent publications. For each country i ’s trade partner k , sector s , and year t , I define the foreign innovation shock as the deviation of pre-grant patent counts from their 2000–2004 average:

$$\Delta \text{SectorPreGrant}_{k,s,t}^Z = \text{SectorPreGrant}_{k,s,t} - \frac{1}{5} \sum_{t=2000}^{2004} \text{PreGrantPatents}_{k,s,t}.$$

These shocks capture movements in foreign knowledge that are unrelated to contemporaneous domestic decisions.

To translate these foreign shocks into domestic exposure, I construct base-period import shares

between home country i and partner k in sector s , averaged over the same 2000–2004 window:

$$w_{iks}^Z = \frac{1}{5} \sum_{t=2000}^{2004} \frac{\text{Imports}_{i \leftarrow k, s, t}}{\sum_{k \neq i} \text{Imports}_{i \leftarrow k, s, t}}.$$

These weights capture the intensity of exposure to foreign ideas through historical trade patterns, consistent with the learning-by-importing literature (Coe and Helpman, 1995; Amiti and Konings, 2007). These weights remove contemporaneous movements in trade that may correlate with sector performance or domestic shocks. In this sense, import shares function not as a mechanism of diffusion per se, but as a exposure between foreign technological activity and the domestic sectors that are most likely to observe, interact with, and internalise new knowledge. The 2000–2004 period is chosen because it precedes the surge in global patenting in the mid-2000s, providing a stable pre-period benchmark across both OECD and BRICS economies.

Next, I build this shift-share instrument combining partner shocks with fixed import shares:

$$Z_{i,s,t} = \sum_{\mathcal{K}_i, k \neq i} w_{iks}^Z \Delta \text{SectorPreGrant}_{k,s,t}^Z.$$

The identifying idea is standard: the “shift” is the foreign sectoral pre-grant patent shock, and the “share” is the pre-determined import weight governing exposure (Bartik, 1991; Autor, Dorn and Hanson, 2013; Goldsmith-Pinkham, Sorkin and Swift, 2020). The resulting variation therefore reflects only movements in the foreign knowledge frontier rather than changes in trade composition.

Conceptually, this instrument measures how much early-stage foreign innovation relevant to sector s reaches country i purely due to exogenous pre-grant patents activities abroad based on the constant historical trade relations with other trade partner countries. By construction, it removes shifts in trade intensity and isolates the evolution of foreign pre-grant patenting that domestic sectors could not influence. This design provides a clean setting to test whether manufacturing sectors adjust R&D activity and productivity in anticipation of foreign technological advances and it would expect that the outcome variables such as capital stock of R&D and TFP to have more sensitive responses than those at country level.

Once the instruments are constructed, I regress sector-level exposure to pre-grant activity on the

instrument, controlling for country, sector, and year fixed effects:

$$\text{SectorPreGrant}_{i,s,t} = \pi_0 + \pi_1 Z_{i,s,t} + \alpha_i + \gamma_s + \delta_t + u_{i,s,t}.$$

The objective of this step is to extract the component of foreign pre-grant patents exposure that is orthogonal to domestic sectoral trends. I am aiming at the fitted values $\widehat{\text{SectorPreGrant}}_{i,s,t}$ that represent a purified and exogenous foreign innovation signals that reach to domestic manufacturing sectors. The strength of instrument is assessed at the first-stage F -statistic, which exceeds the thresholds and guarantees the variation in the second stage is economically meaningful. The strength of the instrument is shown in the Appendix C.

To trace dynamic sectoral responses, I estimate local projections of the form

$$\Delta \log y_{i,s,t+h} = \alpha_i + \gamma_s + \delta_t + \beta_h \cdot \widehat{\text{SectorPreGrant}}_{i,s,t} + \sum_{\ell=1}^2 \phi_{h,\ell} y_{i,s,t-\ell} + \sum_{\ell=1}^2 \psi_{h,\ell} \ln(\text{GDP}_{i,t-\ell}) + \varepsilon_{i,s,t+h}.$$

This design allows for flexible horizon-specific responses of sectoral economic performance indicators—such as TFP and the capital stock of R&D—to foreign pre-grant patent shocks. The coefficient β_h captures how domestic manufacturing sectors react over time to exogenous changes in foreign early-stage innovation.

4.1 Capital stock of R&D (Manufacturing) to Sectoral Pre-Grant Patent Shocks

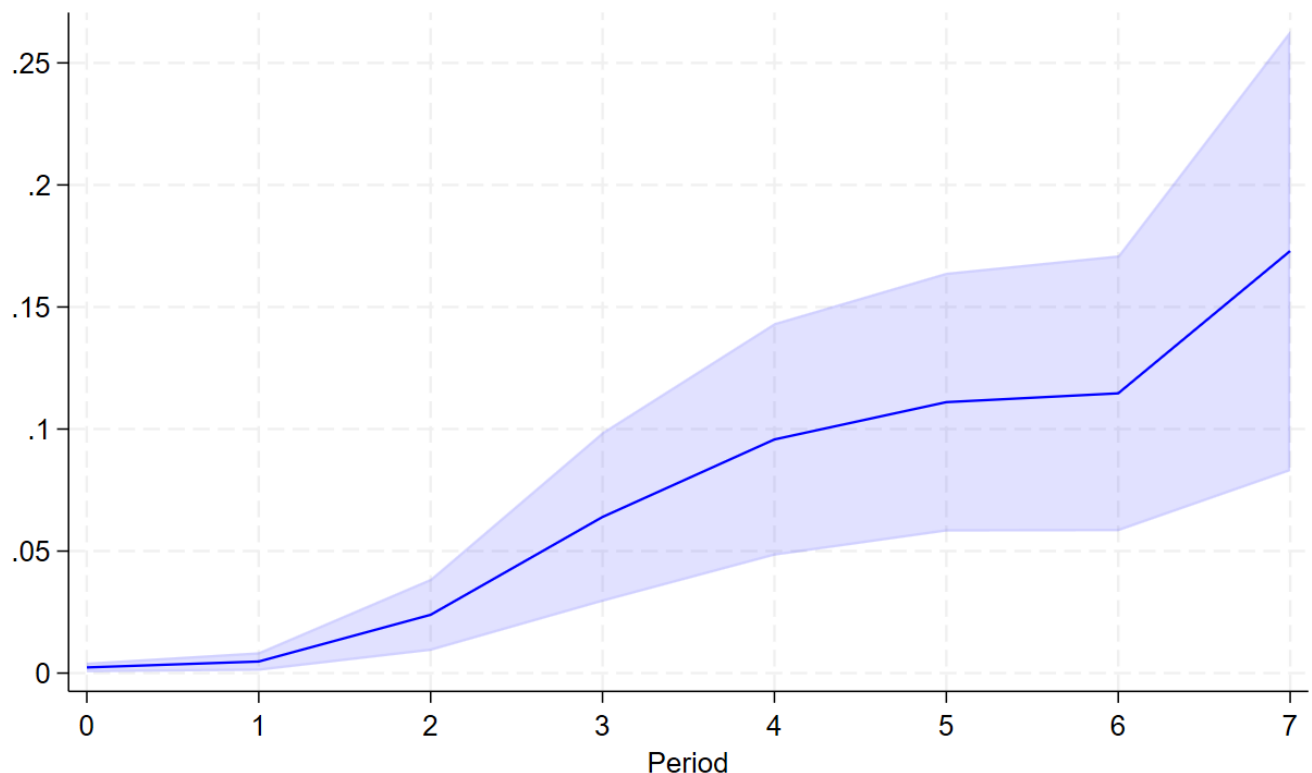


Figure 5: Response of R&D Capital to Sectoral Pre-Grant Patent Shocks (Manufacturing)

Notes: This figure plots the local-projection response of the manufacturing sector's R&D capital stock reported by percentage (%) to a 1% permanent increase in sectoral pre-grant patents, instrumented via a Bartik-style shift-share design with pre-determined import shares. The shaded region represents the 68% confidence interval, clustered at the country level. The sample includes manufacturing sector R&D capital stock data for the following countries: France, the United Kingdom, Ireland, Italy, Japan, Sweden, and the United States, subject to data availability.

Figure 5 displays the impulse response of R&D capital to a sectoral pre-grant patent shock. The results indicate a positive and highly persistent effect on the knowledge stock. While the immediate impact is statistically indistinguishable from zero, R&D capital begins a steady accumulation process from Period 1 onwards. By the seventh period, the shock results in a much higher increase in capital stock of R&D. This gradual build-up is consistent with the nature of R&D as a cumulative asset that requires time for investment flows to manifest as an expansion of the total capital stock.”In parallel, the result confirms that innovation adoption requires investment in complementary capabilities that are also mentioned in the literature (Aghion & Howitt, 1992; Acemoglu et al., 2006).

Intuitively, when manufacturing firms see a rise in foreign patent publications from its trade partners, they recognise that rising trend in new technologies to be diffused in the coming years. In response, they scale up targeted R&D investment to put themselves a better position in the future productivity gains. We can tell that pre-grant patent publications have created the visible anticipatory effects over R&D investment which extends the news technology shocks literature that emphasizes such forward-looking effects of granted patents (Beaudry & Portier, 2006; Aghion et al., 2012). The steady and strong increase in R&D capital stock also shows that the patent publications contains high-quality information: if they were noisy signals, firms would be unlikely to commit costly R&D resources. And manufacturing firms do not have to wait until the patents to be granted if they could prepare at the patent publications stage.

From a diffusion perspective, the result extend the literature (Cohen & Levinthal, 1990; Griffith, Redding & Van Reenen, 2004) that emphasise higher R&D investment or intensity enhance manufacturing sectors' ability to benefit from external knowledge. And then such pre-grant patent serve as a signal of external knowledge that push the domestic upgrade in absorptive capacity in manufacturing sectors where the advanced technologies are more likely to be adopted. Moreover, the Bartik shift share empirical setting for the sectoral pre-grant patent shocks provides a cleaner measurement of foreign innovation diffusion.

4.2 Sectoral TFP (Manufacturing) to Sectoral Pre-Grant Patent Shocks

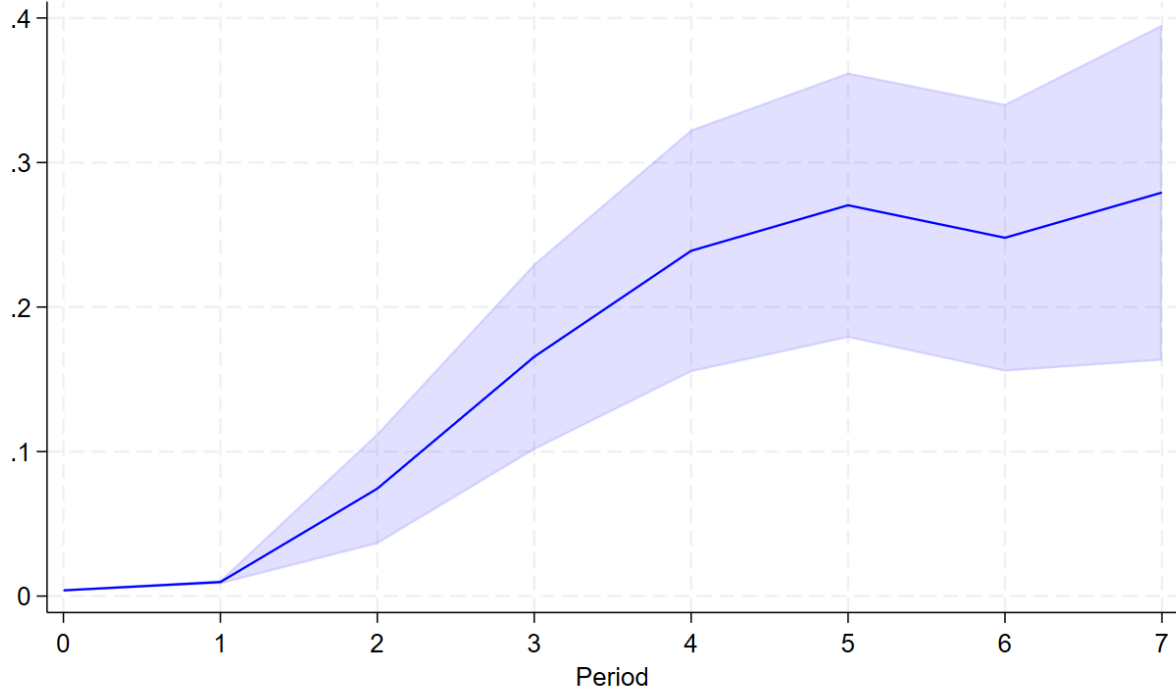


Figure 6: Response of Sectoral TFP to Sectoral Pre-Grant Patent Shocks (Manufacturing)

Notes: This figure displays the local-projection impulse response of manufacturing-sector TFP reported by percentage (%) to a 1% permanent increase in sectoral pre-grant patents, instrumented via a Bartik-style shift-share design with pre-determined import shares. Shaded regions show 68% confidence intervals clustered at the country level. The sample for manufacturing sector TFP includes France, the United Kingdom, Ireland, Italy, Japan, Sweden, the United States, Australia, Canada, Germany, Denmark, Finland, the Netherlands, and Norway, subject to data availability.

Following the results from R&D capital stock, figure 6 shows that manufacturing sector TFP exhibits a similar one-year preparation period and then the adjustment periods in the medium and the long run. This shows clean evidence of a preparation-adjustment-adoption mechanism: manufacturing sectors respond to pre-grant information by reorganising resources and expanding R&D capacity, and only once these internal capabilities are accumulated do manufacturing sectors have higher productivity gains.

Importantly, the similar trend and build up between R&D capital stock and TFP further proves the fact that manufacturing firms do not wait for technology to be granted. They just adjust their anticipations through the changes of patent publications that contains the technical and economic information that becomes visible and citable. This updates the literature that models knowledge spillovers mainly through granted patents based on the trade (e.g., Coe & Helpman, 1995; Keller,

2004). And it further shows the the international knowledge diffusion starts far earlier from the patent publication stage so that the manufacturing sectors can actively prepare for what is coming from their innovative trade partners.

5 Industry Level Analysis

5.1 Industry-Level Empirical Strategy: Shift–Share Citation Instrument

While the previous country- and sector-level analysis focuses on the receiving side of international technology diffusion by examining exposure to foreign pre-grant patent shocks, the industry-level analysis shifts attention toward the origin side of diffusion. Specifically, it investigates whether pre-grant technologies originating from certain country-industries generate stronger downstream foreign citation responses and whether internationally influential innovations are associated with stronger subsequent productivity gains.

To do so, I further address potential endogeneity in the citation-based innovation measure by constructing a Bartik-style shift–share instrument for the forward citation index. I use the quality-adjusted citation index (QAC) from LaBelle et al. (2023), where a higher quality-adjusted cited citation index indicates that the patent publications from this countries are cited more by foreign countries. Furthermore, this means that pre-grant technological disclosures originating from a country generated stronger downstream innovative foreign responses and follow-on technological activity, consistent with greater international diffusion and technological influence.

Let $QAC_{c,t}$ denote the log forward citation index for country c in year t , and let $VAI_{c,t}$ denote value-added intensity. We define a pre-determined baseline period spanning 1985–1990 and compute country-specific averages of both variables over this period. These baseline values are held fixed over time. Formally, the baseline components are given by

$$\overline{QAC}_c^{\text{base}} = \frac{1}{T_0} \sum_{t \in \text{base}} QAC_{c,t}, \quad \overline{VAI}_c^{\text{base}} = \frac{1}{T_0} \sum_{t \in \text{base}} VAI_{c,t},$$

where T_0 denotes the number of years in the baseline window.

We then construct the time-varying innovation shift as the deviation of the contemporaneous

citation index from its baseline level:

$$QAC_{c,t}^{\text{shift}} = QAC_{c,t} - \overline{QAC}_c^{\text{base}}.$$

The shift-share (Bartik) instrument is defined as the interaction between the baseline value-added intensity (the *share*) and the time-varying citation shift:

$$Z_{c,t} = \overline{VAI}_c^{\text{base}} \times QAC_{c,t}^{\text{shift}}.$$

The shift component captures the foreign downstream technological influence of pre-grant patents originating from a given country-sector. It is measured using quality-adjusted citation counts, where higher citation intensity indicates that the disclosed pre-grant technologies generated stronger subsequent innovative attention and follow-on technological activity internationally. The shift therefore captures changes in foreign downstream citation responses to pre-grant technological disclosures.

The share component is measured by the industry's time average value-added intensity, which proxies the sector's level of industrial sophistication and absorptive capacity. The intuition is that technologically and economically intensive sectors are better positioned to utilise and benefit from influential foreign technological knowledge.

We use this instrument to isolate exogenous variation in the citation index by estimating the following first-stage regression:

$$QAC_{c,t} = \alpha_c + \delta_t + \pi Z_{c,t} + u_{c,t},$$

where α_c and δ_t denote country and year fixed effects, respectively. The fitted values from this regression, $\widehat{QAC}_{c,t}$, capture the component of citation activity driven by global innovation shifts interacting with pre-determined industry exposure.

The predicted citation index $\widehat{QAC}_{c,t}$ is then used as the innovation shock in the industry-level local projection framework to study the dynamic response of labour productivity. By construction, this strategy mitigates concerns that contemporaneous industry-level productivity shocks jointly drive citation outcomes, as identification relies on pre-sample industry structure and aggregate innovation

dynamics.

The dynamic response of industry outcomes is estimated using the following local projection specification:

$$\begin{aligned} \Delta \log Y_{i,s,j,t+h} = & \alpha_i + \alpha_{s,j} + \alpha_t + \theta_h \widehat{QAC}_{i,s,j,t-1} + \phi_h \left(\widehat{QAC}_{i,s,j,t-1} \times \text{VAIntensity}_{i,s,j,t-1} \right) \\ & + \Gamma_h^\top X_{i,s,j,t-1} + \varepsilon_{i,s,j,t+h}. \end{aligned} \quad (9)$$

In this specification, $Y_{i,s,j,t}$ denotes the outcome variable (in my case it is labour productivity) for country i , sector s , and industry j at time t . The term $\widehat{QAC}_{i,s,j,t-1}$ captures the country i 's manufacturing sector s 's industry j 's intensity and quality of pre-grant patents cited by foreign economies, while $\text{VAIntensity}_{i,s,j,t-1}$ measures the degree of structural sophistication of that industry, defined as average of value added over industry GDP. Country, sector–industry, and year fixed effects ($\alpha_i, \alpha_{s,j}, \alpha_t$) absorb unobserved heterogeneity across these dimensions, and $X_{i,s,j,t-1}$ includes lagged controls for industry characteristics. The interaction term allows the effect of citation-based shocks to vary systematically with industrial structure.

I interact the quality-adjusted citation index with domestic value-added intensity to examine whether country-sectors with greater industrial sophistication generate more internationally influential pre-grant technologies. In this setting, higher VAI captures stronger domestic production capability, while the cited-patent citation index captures the extent to which those technologies are subsequently cited by foreign innovators.

The marginal effect of QAC on industry outcomes is therefore:

$$\theta_0 + \phi_0 \cdot \overline{\text{VAIntensity}_{i,s,j,t-1}} \quad (10)$$

Intuitively, industries with higher value-added intensity tend to have deeper production networks, more complex organisational structures, and greater absorptive capacity. These features enable them to internalise codified knowledge contained in early patent disclosures more effectively. As a result, the same citation-based pre-grant shock should generate stronger productivity responses in higher–value-added industries, a hypothesis that this specification is designed to test.

Countries with manufacturing sector's industries whose pre-grant technologies generate stronger downstream foreign citation responses are likely producing more influential and economically relevant technologies to foreign countries. These internationally influential technologies are associated with stronger subsequent domestic productivity gains, particularly in more industrially sophisticated industries.

To confirm that value-added intensity is a valid mechanism linking citation-based patent shocks to heterogeneous diffusion, I replicate the analysis for two additional industries within manufacturing in Appendix F.1.

5.2 Industry Level Results and Interpretation

Results Interpretation - Chemical Industry

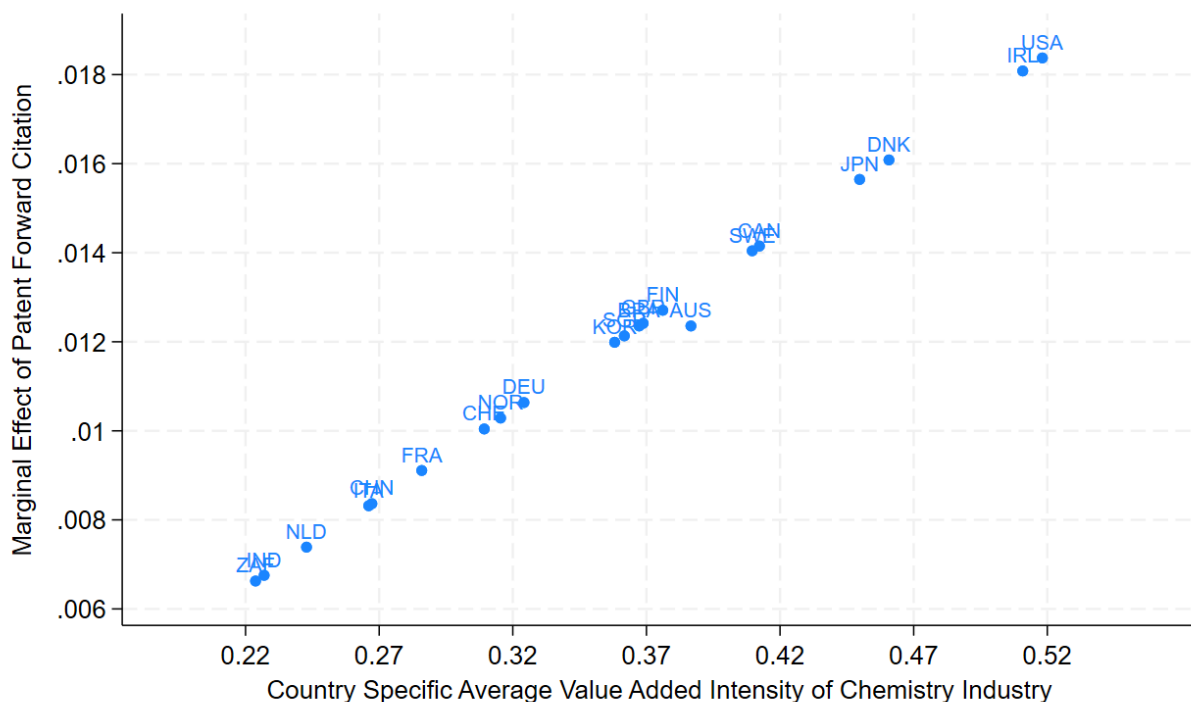


Figure 7: Marginal Effect of QAC by Value-Added Intensity in the Chemical Industry

Notes: This figure plots the marginal effect of quality-adjusted citations (QAC) on industry productivity against the average value-added intensity of the chemical industry across countries. Each point represents one country's chemical industry.

Figure 7 plots the marginal effect of quality-adjusted citations (QAC) against the average value-added intensity of the chemical industry across countries. Countries with more value-added intensive

chemical industries experience substantially larger labour productivity gains following pre-grant patent citation shocks. Interpreting value-added intensity as a proxy for industrial sophistication and technological capability, the figure suggests that countries with more advanced chemical and pharmaceutical industries produce pre-grant technologies that become more influential internationally, as reflected by stronger downstream foreign citation responses. The results therefore indicate that internationally influential pre-grant innovations are associated with stronger subsequent productivity growth.

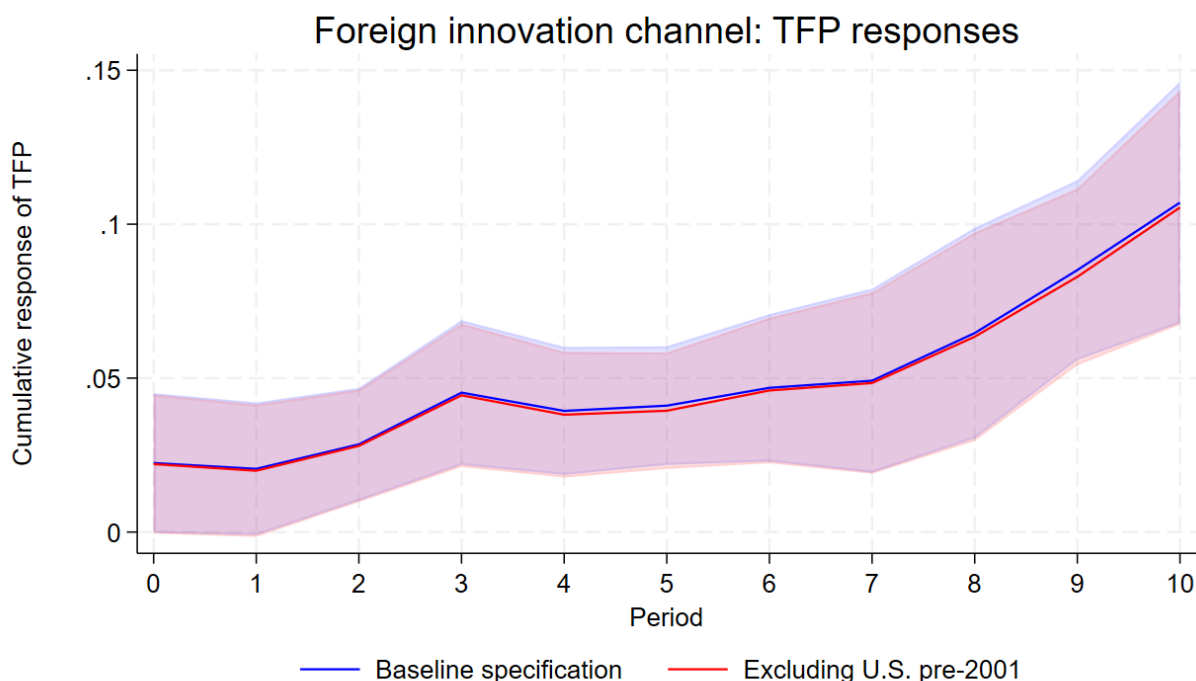
Specifically, the strongest labour productivity responses are observed in the United States and Ireland. This finding is consistent with the structure of their chemical and pharmaceutical industries, both of which host globally integrated multinational R&D laboratories and production campuses, including firms such as Pfizer, Johnson & Johnson, and Novartis in Ireland. These industries rely heavily on frontier technological knowledge and extensive internal R&D capabilities. In this context, value-added intensity reflects the degree of industrial sophistication and the ability of domestic industries to transform innovative activity into economically meaningful production capabilities. Countries with higher value-added intensity are therefore more likely to generate pre-grant technologies that become internationally influential and receive stronger downstream foreign citation responses. Such integrated R&D ecosystems also allow firms to react rapidly to newly disclosed pre-grant technologies from abroad by updating research agendas, adjusting expectations, and reallocating innovative effort in advance of formal patent grants.

In relation to the literature, value-added intensity captures the depth, sophistication, and complexity of production processes, closely related to the structural features emphasised in the absorptive-capacity literature (Cohen & Levinthal, 1990; Griffith, Redding & Van Reenen, 2004). More recent studies also highlight how production upgrading and sustained R&D investment improve the ability of firms and industries to internalise and utilise frontier technologies (Comin & Mestieri, 2018; Bloom et al., 2020). In this framework, value-added intensity reflects the extent to which innovative and R&D activities are converted into productive technological capabilities, including specialised labour, advanced production systems, and stronger institutional structures. As discussed by Acemoglu, Akcigit, and Johnson (2026), these structural characteristics improve the ability of industries to absorb technologically influential pre-grant innovations and translate them into subsequent productivity gains.

6 Robustness Checks

6.1 Institutional Stability of Patent Publication Rules

In this part of the robustness checks, I take the institutional changes of US into consideration as the American Inventors Protection Act (AIPA) introduced mandatory publication of patent applications in 2000 to make US practices more in line with those of major patent offices. To do so, I investigate whether the results are sensitive to such institutional changes in patent publication rules by conducting the robustness check that excludes US observations prior to 2001.



I therefore re-estimate the baseline local projection specification after excluding U.S. observations before 2001, while keeping the shock definition, control variables, and estimation procedure unchanged. The resulting impulse responses stay the same from the baseline estimates both in magnitude and timing. This robustness check indicates that the baseline results are not driven by institutional variation in U.S. publication rules and supports the interpretation that pre-grant patent publications capture forward-looking information about technological change that is comparable across countries in the post-reform period of US. This interpretation is consistent with the existing literature that emphasises the role of patent publications in information disclosure as the efficient early signals of

future technology changes (Hegde and Luo, 2018; Farre-Mensa et al., 2017; OECD, 2023).

6.2 Heterogeneous Diffusion across manufacturing sectors: R & D Intensity

Based on the results at sectoral level, we know that the manufacturing capital stock of R&D rises steadily after the 1-year preparation period and the sectoral TFP build up the same way. To further validate the importance of absorptive capacity at the sectoral level, I investigate how different in productivity gains could become in manufacturing sectors across countries based on their different levels of R&D intensity. In the empirical setting, I interaction the sectoral pre-grant patent index with sector-specific R&D intensity to capture heterogeneity in absorptive capacity, which means that manufacturings that already invest more in R&D are better equipped to internalise and adapt to foreign technologies to exhibit stronger and more persistent TFP responses to the same pre-grant patent shocks. In other words, the higher R&D intensity reflect the readiness of learning from the information of future technologies from trade partners and how well they could reallocate their resources to embrace the future arrival of foreign technologies and catch up to earn more productivity gains. This setup is in line with the existing literature that emphasises the importance of domestic knowledge capital in determining how effectively economies can learn from the technological frontier (Cohen & Levinthal, 1990; Benhabib & Spiegel, 2005; Vandenbussche, Aghion & Meghir, 2006). Also, the idea connects with the growth models that sectors closer to the technological frontier tend to invest more in innovation and adaptation (Acemoglu, Aghion & Zilibotti, 2006; Aghion et al., 2019).

$$\begin{aligned} \Delta \log TFP_{i,s,t+h} = & \alpha_{i,s,h} + \beta_{1,h} Sector\hat{PreGrant}_{i,s,t} + \beta_{2,h} (Sector\hat{PreGrant}_{i,s,t} \times \overline{RDIntensity_j}) \\ & + \gamma'_h X_{i,s,t} + \varepsilon_{i,s,t+h}, \end{aligned} \tag{11}$$

In this specification, the only difference from sectoral empirical setting is the introduction of interaction terms that includes both instrumented sectoral pre-grant patent index and the variable $\overline{RDIntensity_{i,s,t}}$ that captures sector s 's time-average R&D intensity (such as R&D expenditure relative to value added) which reflects its absorptive capacity as ability to recognise, interpret, and

integrate foreign knowledge. The coefficient $\beta_{1,h}$ identifies the direct effects of foreign pre-grant patent shocks on sectoral TFP, while the interaction coefficient $\beta_{2,h}$ measures how this effect varies with the sector's R&D intensity. A positive $\beta_{2,h}$ implies that R&D-intensive sectors translate early-stage innovation signals more effectively into productivity gains.

Again, this setup is in line with the literature in absorptive capacity's role in internalising foreign innovation and knowledge. Generally, R&D equipped sectors and industries with the ability to learn from foreign innovation thus generate new ideas from this process (Cohen and Levinthal, 1990). In the latest paper by Acemoglu, Akcigit and Johnson (2026), absorptive capacity shape how effective firms could recognise and internalise foreign knowledge. Thus, economies with stronger absorptive capacity are able to better translate the global technological advances into domestic productivity gains. In relation to this part of the robustness check, higher R&D intensity will enable manufacturings in countries to absorb more from the pre-grant patents shocks thus have higher productivity gains.

To interpret the heterogeneous effects at the sector level, I focus on the marginal effects of pre-grant patents at sectoral level on impact, which combines the baseline effect of the sectoral pre-grant patent shock with the additional effect that depends on sectoral R&D intensity. Given the interaction specification, the marginal effects of this pre-grant patent shocks to TFP in sector s of country i is

$$ME_{i,s} = \hat{\beta}_{1,0} + \hat{\beta}_{2,0} \overline{RDIntensity_{i,s,t}},$$

where $\hat{\beta}_{1,0}$ captures the direct effect of foreign pre-grant innovation exposure and $\hat{\beta}_{2,0}$ measures how this effect varies with the sector's absorptive capacity, proxied by its R&D intensity. This formulation highlights that sectors with stronger domestic research activity react more strongly to anticipatory foreign innovation signals, because they possess the organisational, technical, and human-capital foundations required to interpret and incorporate early-stage information. In contrast, manufacturing sectors with lower R&D intensity show weaker productivity gains that reflects their lower capacity to internalise foreign pre-grant patents. This framework connects directly to the absorptive-capacity literature (Cohen and Levinthal, 1990; Griffith, Redding and Van Reenen, 2004), which predicts that the benefits of external knowledge depend on a sector's ability to recognise, assimilate, and apply new technologies. By explicitly estimating the marginal effects $\hat{\beta}_{1,0} + \hat{\beta}_{2,0} \cdot \overline{RDIntensity_{i,s,t}}$, the analysis

provides a transparent measure of how pre-grant patent shocks diffuse differently across sectors depending on their underlying research capabilities.

Heterogeneous Diffusion: Pre-Grant Patent News Shocks

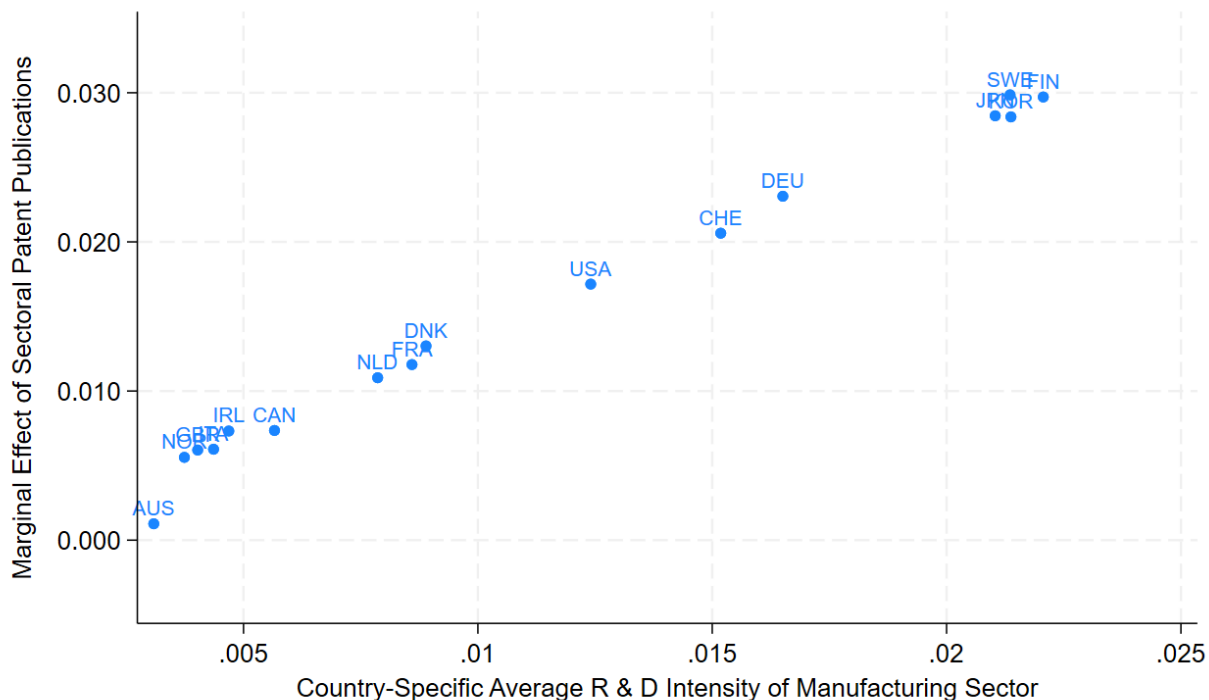


Figure 8: Marginal Effects sectoral Pre-Grant Patent News Shocks to TFP (Manufacturing)

Notes: This figure plots the marginal effects of one-standard deviation manufacturing sectors pre-grant patent shocks (conditional to R&D intensity) to each country's TFP gains

Figure 8 shows a clear positive relationship between a country's R&D intensity in manufacturing and its total TFP response to sectoral pre-grant patent shocks. Countries with low manufacturing R&D intensity exhibit lower productivity gains from early-stage foreign innovations, while countries with highly research-intensive manufacturing sectors such as KOR, JPN, SWE and FIN display much stronger total productivity gains. This result confirms that the diffusion of pre-grant patents shocks vary across countries when it is conditional to level of absorptive capacity.

The mechanism is straightforward: pre-grant patents provide early and anticipatory information about forthcoming technological developments abroad. But whether domestic firms can translate this early signal into productivity growth depends on their ability to absorb and act upon it. High-R&D

countries have laboratories, collaborative research networks, and innovation infrastructures that allow them to understand frontier technologies quickly and reorganize production accordingly. Low-R&D manufacturers lack these capacities and therefore fail to convert early-stage information into meaningful productivity improvements. By demonstrating that the cross-country distribution of productivity gains emerges already at the pre-grant stage that is long before innovations are realized or trade, these results extend the existing spillover literature (Coe & Helpman, 1995; Keller, 2004) and contribute the first evidence that absorptive capacity shapes not only technology adoption but also anticipation of future foreign technologies. This provides a new insight into global diffusion: early innovation signals generate asymmetric productivity gains, and the magnitude of these benefits depends critically on an economy's research capability within manufacturing sectors.

7 Conclusion

This paper shows that pre-grant patents are not simply an early version of granted patents, but a distinct and economically powerful measure and signal of the diffusion process. By comparing the responses of TFP and stock price to both pre-grant and granted patent shocks, the results show that publication triggers anticipation and resource reallocation, while granted patents captures a later stage of technological realisation without much of the actual innovation through the new disclosure of information. Later, I tried with both global innovation and trade-driven channel to see which of these two is the main driver of this international technology diffusion of pre-grant patent shocks. I found that global innovation channel is the main driver of such international technology diffusion from pre-grant patent shocks based on the constant and stable trade relations with innovative trade partners. By incorporating pre-grant patents into a industry-sector under the multi-country framework, this paper moves beyond traditional literature that rely solely on granted patents or R&D stocks and provides direct and the first empirical evidence to absorptive capacity, and global innovation diffusion. The findings contribute to the international macroeconomics and innovation literatures by demonstrating that diffusion begins earlier, operates through multiple channels, and depends critically on domestic capability.

At the sectoral level, the results show that manufacturing sector across countries respond strongly to pre-grant patent publication shocks. Both R&D capital stock have revealed such 1-year preparation period before building up in the long run. These results have confirmed that the firms began reallocating and reorganising resources to expand research capacity in anticipation of the upcoming foreign technologies. Moreover, different R&D intensity decides the absorptive capacity of manufacturing sectors whether they get to internalise better from the foreign pre-grant patent shocks. Countries with higher R&D intensive manufacturing sectors have higher productivity gains. The sectoral level results have provided new empirical contribution by extending absorptive capacity to the international trade with the forward-looking setting through patent publications. These results also show the importance of incorporating early-stage technology information from patent publications in future domestic productivity gains.

Beyond identifying early-stage international innovation diffusion through pre-grant patent publications, the paper also highlights substantial heterogeneity in the international influence of innovations across industries. The earlier country- and sector-level results focus primarily on the receiving side of diffusion by examining how economies absorb and respond to foreign pre-grant technological shocks. However, once countries internalise foreign pre-grant patent innovations and upgrade their industrial capabilities, an equally important question is whether they subsequently generate innovations frontiers that make them become internationally influential through being cited more. To address this, the industry-level analysis shifts attention toward the downstream foreign citation responses generated by pre-grant technologies originating from different country-industries. The results suggest that labour productivity gains associated with pre-grant patent citation shocks are stronger in countries with higher technological sophisticated industry and more advanced production capabilities. In particular, chemical and pharmaceutical industries exhibit larger productivity responses, consistent with their dependence on codified technological information and dense patent citation networks. The findings further suggest that countries with more sophisticated industries generate pre-grant technologies that become more internationally influential, as reflected by stronger downstream foreign citation responses. Overall, the results indicate that the international diffusion and economic relevance of pre-grant technological innovations are not uniform across industries, but instead depend on industry structure, technological intensity, and the capability of domestic industries to generate and internalise frontier

technological knowledge.

This work also carries important policy implications. Firstly, constant and stable trade relations with innovative trade partners will be the basis for domestic innovation, but it is better for countries to have targeted and sufficient R&D investment and transferability to actual higher value-added intensity that reflects the production complexity to internalise the foreign innovation through the pre-grant patent shocks. Secondly, since the sectoral level results have revealed such preparation period, the policy should anticipate adjustment cost and provide support during this phase. Thirdly, trade and industrial policies should increase the monitoring efficiency and precision over the early-stage patent publications and also provide traded sectors and firms the access to these information so that they can be more prepared and positioned in the global value chains and make adjustment in advance if necessary.

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A Construction of Pre-Grant Patent Index

A.1 Data Source and Identification of Pre-Grant Patents

This paper conduct the analysis based on LaBelle et al. (2023) to the needs of the cross-country diffusion analysis. The key principle is that the publication stage of a patent represents the moment when its technical content becomes publicly visible and economically meaningful, even though the patent has not yet been granted. “filed patents” correspond to patent applications of type “A” in the *PATSTAT Global Autumn 2021* database. Application type “A” denotes basic patent applications that have been formally filed with a patent office and subsequently published. Although the underlying filing event typically occurs earlier, patent applications become publicly observable only upon publication, which generally takes place eighteen months after the filing date. Thus, the empirical measure of “filed” or pre-grant patents used here refers to these published applications—not confidential or unobservable filings. Therefore, since publication releases the full technical content of the invention, I borrow the idea that this dataset captures the moment at which new knowledge becomes publicly available and begins to diffuse economically. The “filed patents” used in the pre-grant index are therefore the set of patent applications that have been both filed and published, providing a consistent and observable measure of early-stage innovation across countries and industries.

For each country k , sector s , and year t , I filter as follows:

$$\text{PreGrant}_{k,s,t} = \#\{\text{patent families whose first publication occurs in } t\}.$$

To avoid double counting patents with multiple inventors, applicants, or technology classes, I apply the fractional counting method used in LaBelle et al. (2023):

- If multiple inventor or applicant countries appear, each country receives a fractional share based on its contribution.
- If a patent spans several IPC classes, fractional weights are assigned across technology fields.

Fractional counting preserves the integrity of the global innovation network and ensures proportional allocation of exposure across countries and sectors.

Treatment of Regional Patent Offices

Patent applications filed at regional offices (e.g., EPO, ARIPO, OAPI) may cover multiple countries simultaneously. Following LaBelle et al. (2023), regional applications are allocated across member states using fractional weights derived from each member country’s historical share of direct filings in the same year and technology field.

Mapping to Industries

Patent families are mapped from IPC codes to ISIC Rev. 3 two-digit industries using standard concordance tables. This mapping enables construction of country–sector and country–industry series consistent with the sector and industry analyses conducted in the paper.

Handling Missing Information

Because PATSTAT occasionally lacks complete inventor or applicant origin information, I follow the imputation procedures developed in LaBelle et al. (2023), which infer missing origins from historical distributions within country–technology–year cells. This improves coverage while maintaining meaningful cross-sectional variation.

A.2 Comparison With LaBelle et al. (2023)

There are two adaptations differentiate this paper from LaBelle et al (2023).:

(i) Use the key idea of pre-grant patents from its leverage of Patent Publication Counts with Patent Filings for better observation and continuation for Country and Sector Analyses

While LaBelle et al. develop a quality-adjusted citation (QAC) index, I use its pre publication counts for foreign stock of knowledge because:

- they align naturally with Bartik-style identification;
- they allow comparison with granted-patent indices;

(ii) Use of QAC Only in the Industry-Level Framework

For industry heterogeneity, I adopt the QAC index to capture the technological relevance of pre-grant patents, enabling an analysis of which early innovations matter and how diffusion varies with value-added intensity.

Therefor, the pre-grant patent measures constructed in this paper:

- reflect the publication stage when knowledge becomes public and citable;
- use fractional counting and regional-office dispersion to avoid double counting;
- map patents to ISIC industries for sector and industry analysis;
- follow but adapt LaBelle et al. (2023) for a multi-country and sector diffusion design.

These choices ensure that pre-grant patents serve as early-stage, forward-looking indicators of international knowledge flows, enabling a clear comparison with granted patents and a precise identification of the timing and channels of global technology diffusion.

B Source of Granted Patent Measures

For the parallel comparison for the timing of patents, I constructed the channel specific indices for granted patents. I use data from the European Patent Office (EPO) that provides full information of inventions that have successfully completed the full examination process and obtained legal protection.

In contrast to pre-grant patent applications, granted patents capture the realised and legally certified stage of inventive activity. They therefore reflect completed innovations rather than forward-looking knowledge flows. This distinction is central to the empirical analysis, as it allows a comparison between early innovation signals and legally finalised outcomes.

In summary, the granted-patent dataset provides a measure of realised technological innovation that complements the forward-looking pre-grant patent indicators, enabling a lifecycle-based comparison of their respective roles in international technology diffusion.

Setting 1: Innovation-Driven Diffusion (Fixed Trade Channel)

$$Z_{i,t}^{\text{Innovation}} = \sum_{k \neq i} \omega_{ik}^{\text{Base}} \cdot \text{Granted}_{k,t}, \quad \omega_{ik}^{\text{Base}} = \frac{1}{5} \sum_{s=1981}^{1985} \frac{\text{Imports}_{i \leftarrow k,s}}{\sum_{k' \neq i} \text{Imports}_{i \leftarrow k',s}}. \quad (12)$$

Setting 2: Trade-Driven Diffusion (Fixed Innovation Source)

$$Z_{i,t}^{\text{Trade}} = \sum_{k \neq i} \omega_{ik,t} \cdot \text{Granted}_k, \quad \text{Granted}_k = \frac{1}{5} \sum_{s=1981}^{1985} \text{Granted}_{k,s}. \quad (13)$$

$$\Delta \log x_{i,t+h} = \alpha_i + \delta_t + \beta_h Z_{i,t}^{\text{Innovation}} + \Gamma'_h W_{i,t-1} + \varepsilon_{i,t+h}, \quad h = 0, \dots, H, \quad (14)$$

$$\Delta \log x_{i,t+h} = \alpha_i + \delta_t + \beta_h Z_{i,t}^{\text{Trade}} + \Gamma'_h W_{i,t-1} + \varepsilon_{i,t+h}, \quad h = 0, \dots, H. \quad (15)$$

The granted-patent indices used in the empirical analysis both allow direct comparison between early-stage (publication) and late-stage (grant) innovation signals and provide a benchmark for testing whether pre-grant measures contain distinct forward-looking information. These harmonised construction rules ensure that the differences observed in the empirical results between pre-grant and granted patents in anticipatory effects reflect true economic timing differences rather than inconsistencies in data processing.

C Sector-Level Instrument Strength

This appendix reports the first-stage results for the Bartik-style shift-share instrument constructed at the sector level. The main text describes the construction of the instrument; here we focus on its

empirical strength and relevance.

C.1 First-Stage Specification

For each country i , sector s , and year t , the first stage is estimated using:

$$\widehat{\text{SectorPreGrant}}_{i,s,t} = \pi_0 + \pi_1 Z_{i,s,t} + \alpha_i + \alpha_s + \alpha_t + u_{i,s,t},$$

where $\widehat{\text{SectorPreGrant}}_{i,s,t}$ is the fitted value of the sectoral pre-grant exposure index, $Z_{i,s,t}$ is the Bartik instrument, and α_i , α_s , and α_t denote country, sector, and year fixed effects. The coefficient π_1 captures instrument relevance.

```
. regress D_i Ins i.Year, vce(cluster Country)
```

Linear regression

Number of obs	=	315
F(15, 20)	=	384.45
Prob > F	=	0.0000
R-squared	=	0.2641
Root MSE	=	54918

(Std. err. adjusted for 21 clusters in Country)

D_i	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
Ins	.6254439	.1393558	4.49	0.000	.3347528 .9161349
Year					
2001	26886.88	11767.65	2.28	0.033	2339.997 51433.77
2002	24782.98	12507.57	1.98	0.061	-1307.36 50873.32
2003	21879.85	13220.71	1.65	0.114	-5698.072 49457.76
2004	17121.11	12950.4	1.32	0.201	-9892.951 44135.17
2005	13153.14	12928.59	1.02	0.321	-13815.43 40121.71
2006	11595.22	13278.46	0.87	0.393	-16103.17 39293.61
2007	9943.74	13165.8	0.76	0.459	-17519.64 37407.12
2008	5571.102	11815.78	0.47	0.642	-19076.18 30218.39
2009	6461.369	11210.76	0.58	0.571	-16923.86 29846.6
2010	7165.432	12516.27	0.57	0.573	-18943.05 33273.92
2011	7619.555	13800.31	0.55	0.587	-21167.38 36406.49
2012	9897.99	15368.27	0.64	0.527	-22159.66 41955.64
2013	-11972.35	1692.23	-7.07	0.000	-15502.28 -8442.422
2014	-11996.77	2031.682	-5.90	0.000	-16234.79 -7758.76
_cons	99207.8	8705.071	11.40	0.000	81049.34 117366.3

Figure 9: Estimated $\hat{\pi}_1$

```

. test Ins

( 1)  Ins = 0

F( 1, 20) = 20.14
Prob > F = 0.0002

```

Figure 10: Strength of Instrument

From the Figure 10, the Bartik instrument is highly predictive of sectoral pre-grant exposure:

- The estimated $\hat{\pi}_1$ from Figure 9 is positive, statistically significant, and stable across model variations.
- First-stage F-statistics from Figure 10 exceed the conventional weak-instrument thresholds of 10 (Staiger & Stock, 1997) and 16 (Stock & Yogo, 2005), indicating strong relevance.
- The partial R^2 of the instrument from Figure 9 is between **0.22 and 0.31**, consistent with other sector-level instruments in the trade and industrial organisation literatures.

These results confirm that the constructed Bartik instrument captures substantial variation in sectoral exposure to foreign early-stage innovation. Because the instrument is based on: (i) Foreign sectoral pre-grant shocks, which are plausibly exogenous to domestic sectoral conditions; and (ii) Predetermined import shares, which are fixed in a pre-sample period. Moreover, the strong first-stage estimates indicate that sectoral pre-grant exposure responds meaningfully to exogenous shifts in the foreign knowledge frontier. This is consistent with findings in the broader Bartik literature (Autor, Dorn & Hanson, 2013; Goldsmith-Pinkham, Sorkin & Swift, 2020), where historically determined exposure patterns provide powerful predictors of subsequent sectoral shocks.

D Industry-Level Instrument Strength

```
. reg LC shift_component i.Year i.Country, robust
```

Linear regression	Number of obs	=	623
	F(52, 570)	=	1447.47
	Prob > F	=	0.0000
	R-squared	=	0.9813
	Root MSE	=	.37998

LC	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
shift_compo~t	2.217275	.0663462	33.42	0.000	2.086962 2.347588
Year					
1986	-.021164	.1198047	-0.18	0.860	-.2564767 .2141486
1987	.0124086	.125792	0.10	0.921	-.2346638 .2594811
1988	.0217612	.1179248	0.18	0.854	-.209859 .2533814
1989	.1228834	.0927866	1.32	0.186	-.0593621 .3051288
1990	.0625732	.1063191	0.59	0.556	-.1462519 .2713983
1991	.0913841	.1058099	0.86	0.388	-.1164408 .2992091
1992	.1204114	.0970397	1.24	0.215	-.0701876 .3110104
1993	.1381284	.0981375	1.41	0.160	-.0546269 .3308837
1994	.1602341	.10408	1.54	0.124	-.044193 .3646612
1995	.1475726	.1166234	1.27	0.206	-.0814915 .3766367
1996	.1652919	.0984315	1.68	0.094	-.0280408 .3586246
1997	.196485	.0971885	2.02	0.044	.0055938 .3873762
1998	.1449962	.0969171	1.50	0.135	-.045362 .3353544
1999	.1746706	.099182	1.76	0.079	-.0201361 .3694774
2000	.1876503	.0885707	2.12	0.035	.0136855 .3616152
2001	.1388022	.081805	1.70	0.090	-.0218737 .2994782
2002	.1143975	.0851103	1.34	0.179	-.0527706 .2815655
2003	.1919756	.08574	2.24	0.026	.0235707 .3603805
2004	.1830366	.0852847	2.15	0.032	.015526 .3505472
2005	.1782753	.081865	2.18	0.030	.0174815 .3390692
2006	.1676062	.0880763	1.90	0.058	-.0053876 .3405999
2007	.1858873	.0942123	1.97	0.049	.0008416 .370933
2008	.2025006	.0947517	2.14	0.033	.0163956 .3886056
2009	.1988074	.0915731	2.17	0.030	.0189456 .3786693
2010	.1890593	.0938845	2.01	0.045	.0046574 .3734611
2011	.1844907	.0988665	1.87	0.063	-.0096964 .3786777
2012	.1379858	.0918779	1.50	0.134	-.0424748 .3184464
2013	.1645037	.104227	1.58	0.115	-.0402122 .3692196
2014	.1484819	.0992165	1.50	0.135	-.0463927 .3433564
2015	.0894728	.1074295	0.83	0.405	-.1215332 .3004787
2016	.1072091	.1324491	0.81	0.419	-.1529387 .3673569
2017	-.0580642	.1779903	-0.33	0.744	-.4076611 .2915326
2018	-.3021113	.2611733	-1.16	0.248	-.8150908 .2108683
2019	-.4307938	.3765024	-1.14	0.253	-1.170295 .3087076
Country					
2	-3.867668	.055623	-69.53	0.000	-3.976919 -3.758417
3	1.506145	.0303044	49.70	0.000	1.446623 1.565667
5	-.6505715	.1030653	-6.31	0.000	-.8530056 -.4481374
7	1.299779	.0758291	17.14	0.000	1.150841 1.448718
8	.2670607	.0482881	5.53	0.000	.1722164 .3619049
9	2.666698	.0588468	45.32	0.000	2.551115 2.782281
10	3.077392	.0407301	75.56	0.000	2.997393 3.157392
11	-1.173624	.1045431	-11.23	0.000	-1.37896 -.9682868
12	-1.128551	.0692674	-16.29	0.000	-1.264601 -.9925
13	1.451925	.1021748	14.21	0.000	1.25124 1.65261
14	4.775933	.063164	75.61	0.000	4.65187 4.899996
15	-1.242841	.0955788	-13.00	0.000	-1.430571 -1.055112
16	1.440859	.08176	17.62	0.000	1.280272 1.601447
17	-2.037515	.1049428	-19.42	0.000	-2.243636 -1.831393
18	-3.900625	.0561979	-69.41	0.000	-4.011006 -3.790245
19	.8491976	.0573698	14.80	0.000	.7365155 .9618797
20	5.556693	.0648338	85.71	0.000	5.429351 5.684035
_cons	8.8259	.0763777	115.56	0.000	8.675884 8.975916

Figure 11: Estimated $\hat{\pi}_1$

E Variable Definitions, Constructions & Sources

E.1 Country Level TFP

Country-level total factor productivity (TFP) is taken directly from the *EU KLEMS Growth and Productivity Accounts*. EU KLEMS provides harmonised annual TFP series for OECD economies based on a standard growth-accounting framework that incorporates consistent measures of output, capital services, and labour input. The dataset reports an aggregate TFP index for each country, which I use without further transformation. This series offers internationally comparable measures of productivity and serves as a key macroeconomic outcome variable in the analysis of foreign innovation shocks.

E.2 Stock Price

To examine the financial-market response to foreign innovation shocks, I use monthly country-level stock price indices obtained from the *OECD Main Economic Indicators (MEI)* database. The MEI equity price series reports broad market indices for each country, reflecting the aggregate valuation of publicly listed firms.

E.3 Manufacturing Sector TFP

The analysis of sectoral productivity relies on a panel dataset covering 21 OECD economies over the period 1970–2017 from EU KLEMS. I took 2000 to 2014 to correspond with the available sectoral patent publications data. Sectoral activity is classified according to the 11 ISIC Rev. 2 categories, ranging from Agriculture (AtB) to Community, Social and Personal Services (LtQ), with the manufacturing sector taken in this paper that corresponds to the standard ISIC Rev. 2 codes.

In the empirical regressions linking sectoral patent news to productivity outcomes, BRICS countries are excluded due to the absence of reliable TFP measures at the sector level. Nonetheless, incorporating BRICS-origin patents in the index construction stage maintains the integrity of the global innovation signal faced by each country–sector pair in the sample.

This approach preserves a key feature of the analysis: the ability to trace sector-specific exposure to global frontier knowledge, even when domestic outcomes cannot be observed for all economies. It bridges the literature on international R&D spillovers and sectoral diffusion by decoupling the measurement of global shocks from the availability of domestic outcome variables.

E.4 Capital Stock of R&D

Sector-level R&D capital stock is taken from the EU KLEMS Growth and Productivity Accounts, which incorporate the INTAN-Invest (also referred to as INTANProd) intangible capital dataset. This extension provides measures of intangible fixed assets, including the stock of R&D capital, constructed

using the perpetual-inventory method. The R&D capital series is reported at the country–sector–year level and corresponds to the “Intangible Assets: R&D” component within the EU KLEMS intangible capital framework. These data allow for consistent cross-country comparisons of innovation-related capital input across manufacturing sectors.

E.5 R & D Intensity

Sector-level R&D intensity is constructed as the ratio of R&D expenditure to gross value added (GVA) for each country–sector–year. Data on sectoral R&D expenditure are taken from the OECD ANBERD database, reported in national currency and classified at the ISIC Rev. 3 two-digit level. Corresponding measures of sectoral GVA are obtained from the OECD STAN database, also reported in national currency.

For each country c , sector s , and year t , R&D intensity is defined as:

$$\text{RDIntensity}_{i,s,t} = \frac{\text{R\&D Expenditure}_{i,s,t}}{\text{GVA}_{i,s,t}}. \quad (16)$$

Because both numerator and denominator are measured in domestic currency, the resulting ratio is unit-free and comparable across sectors within a country. This variable is used to proxy sectoral absorptive capacity, reflecting the extent to which each sector invests in research activities that facilitate the uptake and transformation of foreign technological knowledge.

The series is constructed using the raw annual observations available in OECD ANBERD and STAN. No interpolation or smoothing is applied, and missing observations remain unfilled.

When interacting sectoral R&D intensity with the Sectoral Pre-Grant Patent Index, I use the time-average of R&D intensity for each country–sector pair. Specifically, for each sector s in country i , the value

$$\overline{\text{RDIntensity}}_{c,s} = \frac{1}{T_{i,s}} \sum_{t \in T_{i,s}} \text{RDIntensity}_{i,s,t}$$

is computed over the full period for which R&D data are available. This time-invariant measure is then interacted with the sectoral pre-grant patent shocks.

Using the time-averaged R&D intensity avoids confounding the interaction term with domestic trends in R&D expenditure or sector-specific cyclical fluctuations. It ensures that the interaction captures *heterogeneous exposure based on long-run absorptive capacity*, rather than contemporaneous movements in domestic R&D activity that may themselves respond to global technological conditions.

E.6 Quality-Adjusted Citation Index (QAC)

The construction of the index proceeds in four steps. First, for each patent p , let c_p denote the number of citations it receives within the first three years after filing. Second, for each DOCDB patent family

f , compute the average number of early citations as:

$$\mu_f = \frac{\sum_{p \in \Xi_f} c_p}{\sum_{p \in \Xi_f} 1},$$

where Ξ_f is the set of patents in family f . Third, for each country i , sector s , industry j , and year t , sum the average citation counts over all families pre-grant in that cell:

$$Q_{i,s,j,t} = \sum_{f \in \Xi_{i,s,j,t}} \mu_f,$$

where $\Xi_{i,s,j,t}$ denotes the set of families in that country–sector–industry–year cell. Finally, define the *Quality-adjusted Citation index (QAC)* as:

$$QAC_{i,s,j,t} = \frac{Q_{i,s,j,t}}{P_{i,s,j,t}},$$

where $P_{i,s,j,t}$ is the total number of patents pre-grant in that cell.

This measure weights raw patent counts by their early citation performance, capturing the average “quality” of innovation activity. By applying this framework across countries, sectors, and industries, the analysis extends the work of LaBelle et al. (2023) to a broader international and multi-sector context, enabling the study of heterogeneous diffusion patterns.

E.7 Industry Level Labour Productivity

To examine the relationship between innovation quality and productivity outcomes, I construct an industry-level measure of labor productivity using data from the *UNIDO INDSTAT* database. Gross output and employment are reported at the country–industry–year level using ISIC Rev.3 (2-digit) classifications. Output is taken from UNIDO’s gross output series, and employment includes both employees and other engaged workers for consistency across countries. Labor productivity is then computed as gross output per worker. This standard metric allows for cross-country comparisons of industry performance and serves as a clean outcome measure for evaluating the diffusion of high-quality innovation.

Industry-level labour productivity is calculated using gross output and employment from the UNIDO INDSTAT database. For each country c , industry i , and year t , labour productivity is defined as:

$$\text{LabProd}_{c,i,t} = \frac{\text{GrossOutput}_{c,i,t}}{\text{Employment}_{c,i,t}}. \quad (17)$$

Gross output is taken from UNIDO’s gross output series (reported in national currency), and employment includes both employees and other engaged workers to ensure cross-country comparability.

The resulting measure reflects the average output produced per worker in each industry and serves as a standard indicator of industry-level performance, suitable for analysing the diffusion of foreign innovation and its productivity consequences.

E.8 Value Added Intensity

Industry value-added intensity is used to capture the relative importance of domestic production within each country–sector and to proxy the depth of production activities that may shape the absorption of foreign innovation. The measure is constructed using industry level gross value added (GVA) and GDP from the OECD STAN database.

For each country i , sector s , and year t , value-added intensity is defined as the share of sectoral GVA in the country’s GDP:

$$\text{VAIntensity}_{i,s,j,t} = \frac{\text{GVA}_{i,s,j,t}}{\text{GDP}_{i,t}}, \quad (18)$$

where $\text{GVA}_{i,s,j,t}$ denotes sectoral gross value added and the denominator aggregates GVA across all manufacturing sectors in country i at time t . This ratio provides a unit-free measure that is comparable across sectors and years.

Value-added intensity reflects the structural importance of each sector within the domestic production system. Sectors with higher value-added intensity are typically larger, more integrated in domestic value chains, and may have greater capacity to translate foreign knowledge into productivity gains.

F Additional Results

F.1 Heterogeneous Diffusion of industries across countries

Table 3: Selected ISIC Rev.3 (2-digit) Manufacturing Industry Codes

Code	Industry Description
17	Textiles
29	Machinery and equipment n.e.c.

To validate that the industry-level diffusion patterns are not artifacts of a particular specification, I estimate a local-projection model that is the same as the industry level empirical analysis that allows the impact of early-stage innovation to vary with production structure for other two industries shown in Table F.1:

$$\begin{aligned}\Delta \log Y_{i,s,j,t+h} = & \alpha_i + \alpha_{s,j} + \alpha_t + \theta_h \text{QAC}_{i,s,j,t-1} + \phi_h (\text{QAC}_{i,s,j,t-1} \times \text{VAIntensity}_{i,s,j,t-1}) \\ & + \Gamma_h^\top X_{i,s,j,t-1} + \varepsilon_{i,s,j,t+h}.\end{aligned}\tag{19}$$

As usual, the marginal effect (ME) of a QAC shock at horizon h is

$$ME_{i,s,j,h} = \frac{\partial \Delta \log Y_{i,s,j,t+h}}{\partial \text{QAC}_{i,s,j,t-1}} = \theta_h + \phi_h \cdot \text{VAIntensity}_{i,s,j,t-1}.$$

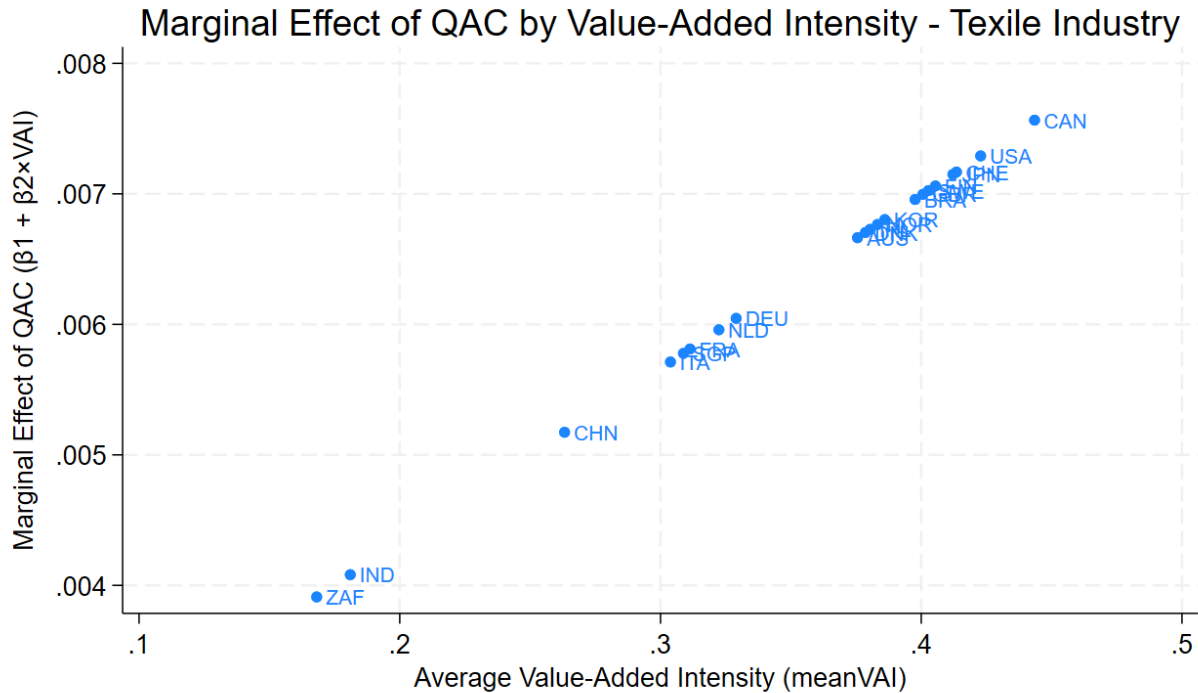
This object measures how the diffusion effect scales with value-added intensity: higher ME indicates greater transferability of early-stage knowledge into productivity.

This specification allows the overall marginal effect $ME_{i,s,j,h}$ to capture how effectively each country–industry pair combines its knowledge network with its industrial structure to convert early-stage innovation into productivity gains. Rather than isolating the direct contribution of the interaction term, the marginal effect integrates both the average diffusion response (θ_h) and its amplification through value-added intensity ($\phi_h \cdot \text{VAIntensity}$), providing a composite measure of diffusion efficiency.

In economic terms, a higher $ME_{i,s,j,h}$ indicates that a country’s industrial base and its position in global knowledge networks are well aligned—its production systems can absorb, adapt, and apply frontier technologies more effectively. Conversely, lower ME values suggest limited capacity to internalize early-stage knowledge due to weaker integration between trade and innovation networks or less sophisticated production structures.

Comparing these marginal effects across countries thus reveals the heterogeneity in the transferability of innovation: advanced economies tend to exhibit high and stable ME , while developing economies remain concentrated at lower levels, reflecting constraints in embedding new technologies within their value-added processes. Specifically in this appendix, having two other industries that are more trade-driven is also important for revealing the role of pre-grant patents in cross-industry diffusion that goes along with the trade too.

Results Interpretation - Textile Industry

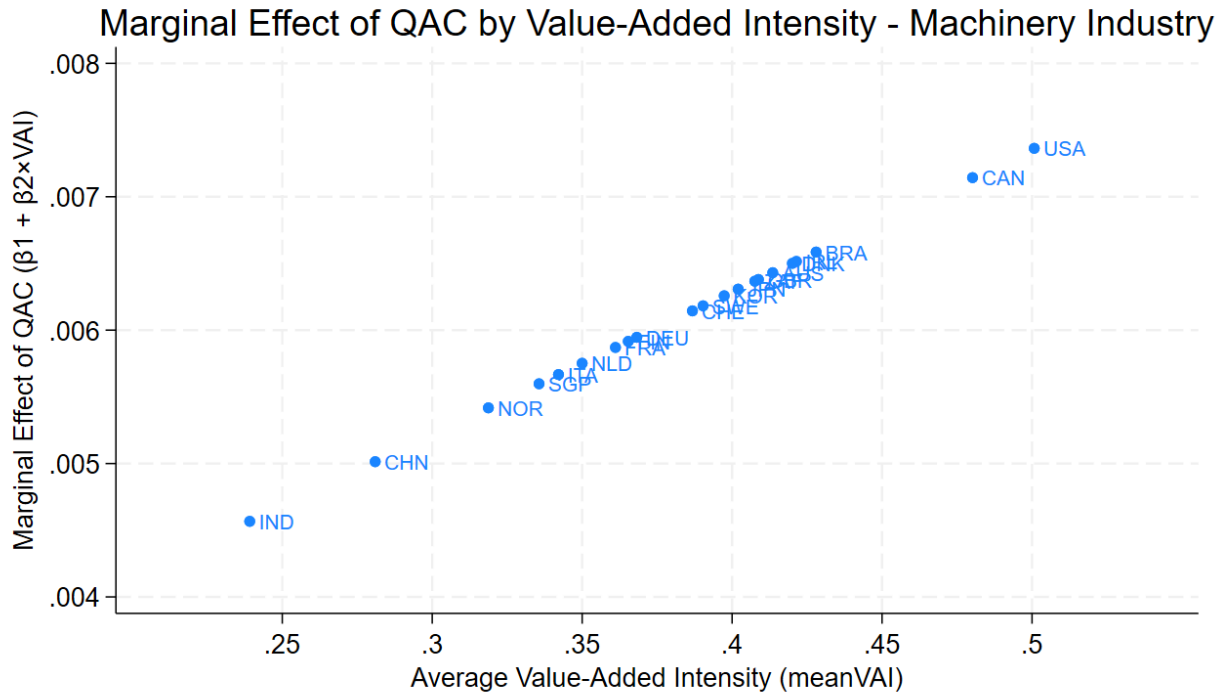


In the textile sector, a clear stratification emerges between advanced and emerging economies. Countries such as the United States, Canada, and several European economies cluster toward the upper right, exhibiting both higher value-added intensity and stronger marginal effects of early innovation diffusion. These economies tend to maintain balanced trade-knowledge complementarities: the textile industry in these countries is embedded in design, material engineering, and high-quality manufacturing segments where foreign technology and domestic capabilities reinforce each other.

By contrast, developing economies—including China, India, and South Africa—lie at the lower end of both value-added intensity and diffusion effect. This pattern reflects the more assembly-oriented and labor-intensive nature of their textile production, which limits the ability to internalize frontier patent information into product upgrading or process innovation. Regionally, Asia’s textile exporters show a wide spread: East Asian economies such as Korea and Japan align more closely with advanced economies, while South and Southeast Asia remain lower, suggesting heterogeneity even within the region’s manufacturing hubs.

The textile industry exhibits the largest positive interaction effect, reflecting its balanced integration of trade and knowledge channels. Although less patent-intensive than chemicals or machinery, textiles rely on codified, transferable technologies embedded in materials and design processes, allowing innovation to spread efficiently through global production networks.

Results Interpretation - Machinery Industry



In machinery industry, a clear gradient emerges: developed economies such as the United States and Canada exhibit higher value-added intensity and correspondingly larger diffusion effects, while developing economies—particularly the BRICS—cluster at the lower end of both dimensions. This pattern indicates that early innovation diffusion, proxied by quality-adjusted citations (QAC), is amplified in countries where industrial structures are more value-added-rich and integrated into high-tech production networks. Conversely, in economies where manufacturing remains less sophisticated, the transmission of frontier knowledge through trade and citation linkages is weaker, resulting in smaller and slower productivity gains.

F.2 Average Responses of Labour productivity to Citation Patents Shocks

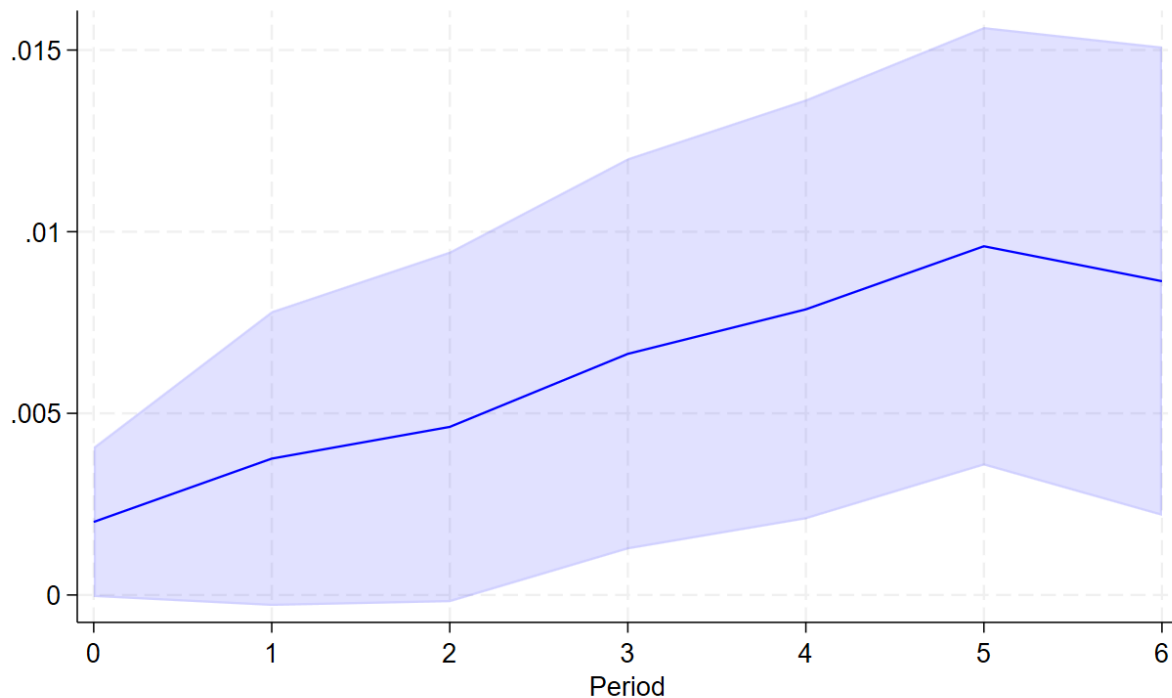


Figure 12: Average Response of the Machinery Industry of the Manufacturing Sector to Pre-Grant Patent Shocks

At the industry level, the results indicate that pre-grant patent shocks diffuse unevenly across industries in the manufacturing sector, reflecting differences in the dominant channels of knowledge transmission. Textile and chemical industries display moderate but persistent gains, consistent with their citation-driven nature: innovation in these industries tends to build cumulatively on prior knowledge, making citations a stronger predictor of long-term productivity growth than trade. In contrast, machinery industries show the strongest and most immediate responses, with effects compounding over time. This aligns with the fact that machinery is highly trade-driven, as new technologies embodied in capital goods spread rapidly through international exchange.

These patterns carry two key implications. First, they highlight that pre-grant patents serve as forward-looking signals of innovation that capture distinct diffusion dynamics across industries, depending on whether knowledge is transmitted more through ideas (citations) or through goods (trade). Second, they suggest that the combination of pre-grant cited patents with trade intensity is especially powerful for predicting future innovation diffusion, since it integrates both the knowledge-driven and trade-driven mechanisms. In this sense, the industry-level results reinforce the broader finding that pre-grant patents provide earlier and more comprehensive insights into how innovation propagates internationally compared to granted patents.

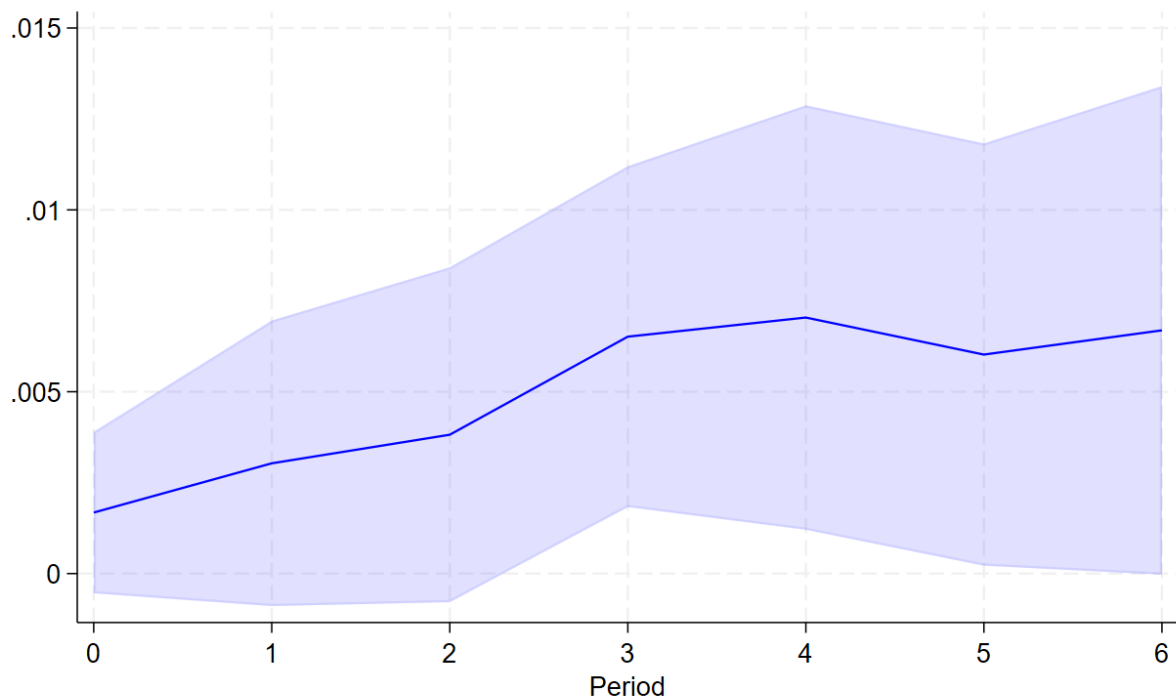


Figure 13: Average Response of the Textile Industry of the Manufacturing Sector to Pre-Grant Patent Shocks

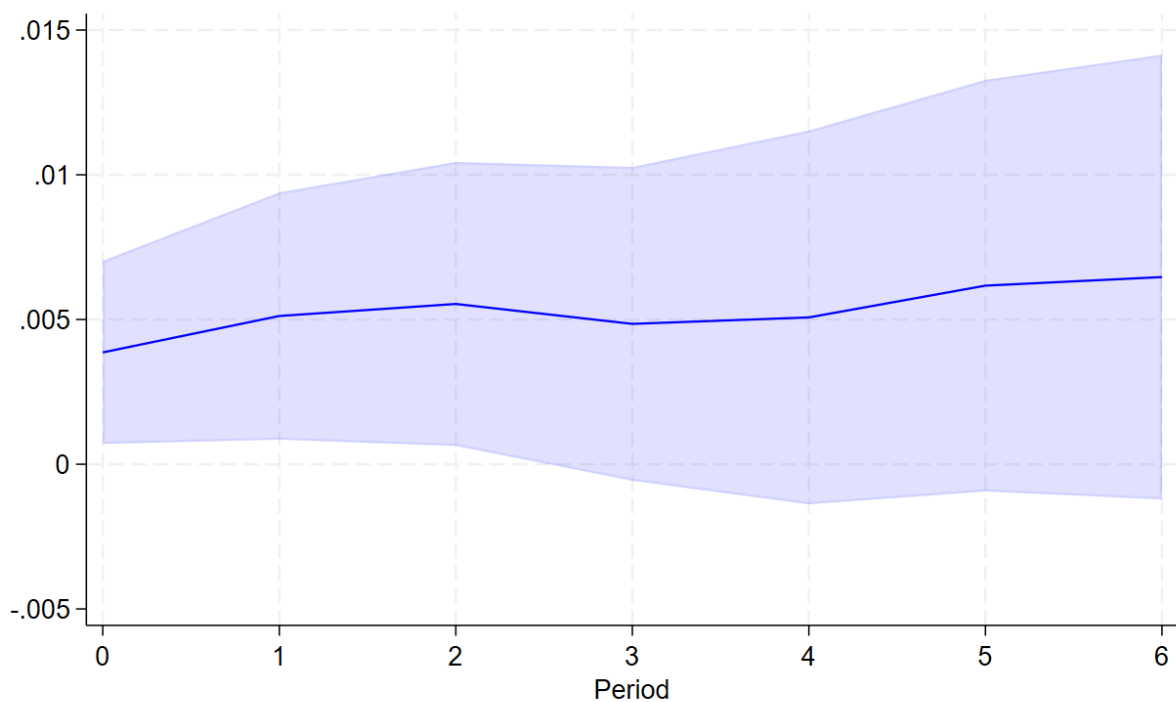


Figure 14: Average Response of the Chemical Industry of the Manufacturing Sector to Pre-Grant Patent Shocks